The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Objective. Temporal-interference (TI) stimulation requires three stages in series: the kilohertz field must reach a nonlinear membrane (access), the membrane must generate the difference frequency absent from the field (detection), and the circuit must amplify that weak component (amplification). We build a mesoscale model. **Approach.** Following single-neuron TI models, we represent access by a short effective membrane time. We analyze detection and amplification in a Jansen–Rit column, an exact mean field of quadratic integrate-and-fire neurons, and its spiking network. **Main results.** For weak fields the detector is quadratic: the beat is proportional to the product of the local carrier amplitudes, and a signed steady source to their summed squares. A static rate curve and the exact mean field give the same product law; the mean field adds frequency dependence when membranes are slow relative to input. In a realistic head with a published hippocampal montage, changing the current ratio moves the envelope and mean-square foci in opposite directions and only rescales the product map. Near a Hopf bifurcation the circuit amplifies the beat as the inverse of the modal damping; with the Jansen–Rit operating point held fixed, synaptic coupling sets this gain independently of detection. The amplified response appears mainly in envelope-locked timing, with little change in mean rate. At stronger fields the steady shift alone can drive a gamma circuit across its oscillation threshold, as can an unmodulated carrier of equal power. **Significance.** The model expresses TI efficacy as the product of three separately measurable factors. For weak fields, dose follows the product of carrier amplitudes rather than the envelope minimum. Under uniform conductivity, access, and detection, the steady drive has its largest magnitude at compartment boundaries. TI shares its circuit amplifier with direct low-frequency stimulation. Carrier sweeps, independent-amplitude tests, beat sweeps across states, and equal-power controls each test one stage.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

Contents

1 Introduction **3**

2 Theory **4**
2.1 A population rate curve supplies the nonlinearity 4
2.2 From applied field to a squared-envelope beat component 5
2.3 Envelope oscillations and spatial weighting 6
2.4 Detector laws and harmonic signatures 7
2.5 The steady source has a state-dependent sign 8
2.6 Amplification: the synaptic low-pass and the network resonance 9
2.7 Carrier access depends on the effective membrane response 10
2.8 Operating-point dependence 12
2.9 The exact mean field extends the static detector 12

3 Methods **13**
3.1 Stimulation waveforms and matched-source controls 13
3.2 Amplitude scaling and harmonic measurements 14
3.3 Head model and field computation 14
3.4 Single-column Jansen–Rit: integration, lock-in, and controls 16
3.5 Exact next-generation mean field (NMM2 PING) 17
3.6 QIF spiking network (microscale of NMM2) 19
3.7 Coupling sweeps, timing, and the tACS control (Jansen–Rit) 20
3.8 Vocabulary: forced response, entrainment, and gating 20

4 Results **20**
4.1 The JR column generates and resonantly amplifies the beat 20
4.2 Carrier independence and the kHz roll-off 21
4.3 The nonlinearity is the detector: the σ'' curvature law and its controls 22
4.4 Realistic-head comparison of envelope and quadratic maps 22
4.5 Coupling sets the gain at fixed detection 25
4.6 The resonance amplifies the envelope-locked modulation more than the mean rate 25
4.7 TI and tACS share the linear amplifier 26
4.8 Exact mean-field confirmation: the NMM2 PING gamma generator 26
4.9 QIF responses distinguish beat modulation from field-induced gamma 27

5 Discussion **30**
5.1 Cellular access and mixing require separate calibration 30
5.2 Detector laws predict different spatial responses 30
5.3 Field-induced oscillations and resonant amplification 31
5.4 Efficiency depends on the generated drive 32
5.5 Limits of the population description 32
5.6 Falsifiable predictions and tests 33

6 Conclusion **35**

A Notation and units **42**

1 Introduction

Temporal-interference (TI) stimulation was proposed to reach deep brain targets while reducing stimulation of surrounding tissue [1]. It applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs, typically with $f_1 \sim 1$–2 kHz and $\Delta f \sim 1$–100 Hz. Where the fields overlap, their superposition is an amplitude-modulated (AM) signal with an envelope at Δf. Electrode currents can steer the region of largest envelope modulation. Whether this also localizes the neural response depends on how neurons transduce the waveform.

The difference frequency is absent from the input itself:

$$s(t) = \varepsilon_1 \cos(\omega_1 t) + \varepsilon_2 \cos(\omega_2 t), \qquad \omega_{1,2} = 2\pi f_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta f. \tag{1}$$

Here $s(t)$ is the local membrane polarization, t is time, and $\varepsilon_{1,2}$ are its tone amplitudes in mV, produced by coupling of the applied field to the membrane; Ω is the beat angular frequency, and $f_c = (f_1 + f_2)/2$ is the carrier frequency. The spectrum contains only the two applied lines. A linear, time-invariant system can attenuate or phase-shift them but cannot create a new one. Passive membrane filtering therefore cannot demodulate TI [2]; a nonlinear element must mix the carriers into the low-frequency band, as in AM reception [3].

The mechanism has three physical requirements. *Access* is the conversion of the extracellular field into local membrane polarization. *Detection*, or demodulation, is the nonlinear generation of the difference-frequency component from those polarizations. *Amplification* is the response of the neuron and its recurrent circuit to the generated component. The same cellular structures can contribute to more than one stage, but strong coupling, effective mixing, and large resonant gain are different properties.

Somatic charging attenuates kHz fields, with the response also depending on cable geometry and field coupling [4, 5]. Esmaeilpour et al. [6] already modeled TI sensitivity using a short, axon-like membrane time. Cohen et al.'s conducting periaxonal space separates nodal and internodal voltage dynamics and supports saltatory conduction [7]; nodes and the sodium-channel-rich axon initial segment (AIS) are candidate access sites [8]. We represent this route by a short effective membrane time, whose field-to-polarization transfer requires calibration.

Detection requires that this polarization encounter a nonlinearity. Mirzakhalili et al. [2] demonstrated subthreshold nodal mixing in a modified McIntyre–Richardson–Grill (MRG) myelinated-axon model with Hodgkin–Huxley ion-channel dynamics. Cortical-neuron models also show axonal excitation and conduction block under stronger fields [9]. Luff et al. [10] directly observed subthreshold difference-frequency oscillations in individual neurons, including carriers near 5 kHz. Mixing persisted during synaptic blockade but was suppressed by blocking tetrodotoxin-sensitive sodium channels. These measurements establish a cellular detector without locating it within a specific compartment. In a population description, smooth cellular contributions can be combined into an effective second-order response; a static rate curve realizes that response through its curvature.

Amplification determines whether the generated beat appreciably changes activity. Intrinsic currents, including the hyperpolarization-activated current I_h and M-type potassium current I_M, can tune a single neuron [11], as in the cell- and state-dependent hippocampal TI responses of de Palma Aristides et al. [12]. Recurrent excitatory–inhibitory circuits supply another resonance [13, 14, 15]. In slices, pyramidal and parvalbumin (PV) neurons respond differently, and most pyramidal TI responses vanish under synaptic blockade [16]; simulations also find network mediation [17, 6]. Reviews summarize these mechanisms [18, 19]. At human-comparable fields, primate recordings show weaker effects than direct low-frequency stimulation, expressed mainly in spike timing rather than mean rate [20]. These findings motivate a model that separates the cellular beat source from its recurrent amplification.

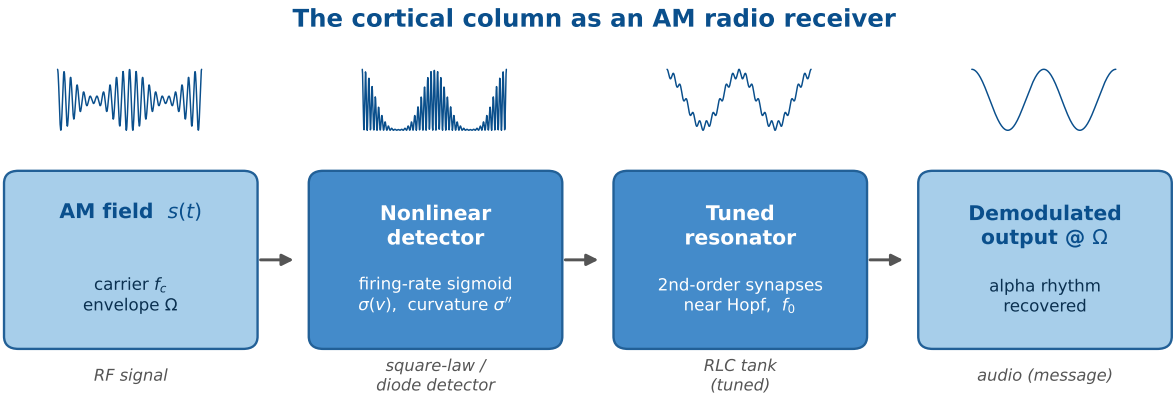


Figure 1: The cortical column as a tuned receiver. Access converts the applied field into the local polarization entering the detector. The nonlinear detector generates the beat at angular frequency Ω from a carrier near f_c; a recurrent network amplifies it near its natural frequency f_0. The diagram shows the static-sigmoid realization. In a conventional AM receiver, the tuned circuit selects the carrier before detection [3]; here the resonator selects the low-frequency output.

We focus on weak stimulation-induced changes around ongoing activity, which may already include spiking. Direct carrier-evoked spiking and conduction block [21, 22, 23] lie outside this expansion; the finite-amplitude network examples are identified separately.

Our mesoscale model separates the cellular beat source from recurrent gain. A Jansen–Rit column establishes the weak-input product law; an exact QIF mean field and its spiking network test how membrane dynamics and recurrent activity modify the response. A laminar extension adds two resonant bands at low carriers. Fast population relaxation gives a static transfer curve with dynamic synapses; slower relaxation requires a dynamic detector. Related neural-mass models describe endogenous cross-frequency coupling and nested amplitude modulation [24, 25]; an external field additionally requires field-to-cell access. Each stage has a separate test: carrier sweeps of local polarization test access, independent variation of the carrier amplitudes tests detection, and beat sweeps across states test amplification. Equal-power unmodulated-carrier controls distinguish the steady shift from the beat response.

2 Theory

2.1 A population rate curve supplies the nonlinearity

A neural mass model describes a population through averaged variables [26, 27]. In the Jansen–Rit (JR) family, a linear synaptic filter converts presynaptic firing rate $r(t)$ into postsynaptic potential $u_{\mathrm{PSP}}(t)$:

$$\ddot{u}_{\mathrm{PSP}} + 2a\dot{u}_{\mathrm{PSP}} + a^2 u_{\mathrm{PSP}} = Aa\, r(t),$$

with inverse synaptic time a (s^{-1}) and PSP gain A (mV); inhibitory synapses use b and B in the same respective units. Summed postsynaptic potentials give the mean input v, which a static transfer curve converts to firing rate [28],

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}}. \tag{2}$$

Here $2e_0$ is the maximum rate, v_0 the inflection voltage, and ρ the inverse voltage scale (s^{-1}, mV, and mV^{-1}, respectively). Primes denote derivatives with respect to voltage. Nonzero curvature makes this curve a mixer, while its slope sets the local gain of the recurrent circuit.

The JR column couples pyramidal, excitatory-interneuron, and inhibitory-interneuron populations, with field input to the pyramidal sigmoid and local field potential (LFP) proxy $v = y_1 - y_2$, where y_1 and y_2 are the excitatory and inhibitory postsynaptic potentials onto the pyramidal population. Its external input rate is p (s^{-1}), and C is the dimensionless connectivity scale.

2.2 From applied field to a squared-envelope beat component

We neglect electromagnetic propagation and induction in the applied field $E(t)$ (V/m), while retaining the membrane's finite charging response. Let $s(t)$ be the polarization (mV) after the cellular access response; the static population model reads

$$r(t) = \sigma\big(v(t) + s(t)\big), \qquad E(t) = E_1 \cos\omega_1 t + E_2 \cos\omega_2 t. \tag{3}$$

Soma, dendrites, AIS, nodes, and terminals have different couplings and response times. We represent their contributions as parallel inputs. For nearby carriers centered at $\omega_c = 2\pi f_c$,

$$s(t) \simeq \sum_k \boldsymbol{\lambda}_f^k(\omega_c)\cdot\mathbf{E}(t), \qquad \boldsymbol{\lambda}_f^k(\omega) = \boldsymbol{\lambda}_0^k\,|H_k(\omega)|, \tag{4}$$

where $\boldsymbol{\lambda}_0^k$ is the static polarization vector of compartment k (a length), and H_k its access filter [26, 29]. Equation (4) uses amplitude gains and assumes aligned compartment phases; a general sum requires the complex transfers, including their relative phases. The access filter is included in the supplied polarization or computed by the membrane equation when its input is current; applying both would count the same attenuation twice.

For a dominant entry point, define $\lambda_f(\omega) = \lambda_0|H(\omega)|$. Then $\varepsilon_j = \lambda_f(\omega_j)E_j$ denotes the polarization reaching the nonlinearity. We first hold this local polarization fixed to isolate mixing and recurrent amplification. At fixed applied field, its frequency dependence follows H. Fast axonal compartments motivate the direct-coupling approximation $\lambda_f \approx \lambda_0$; their coupling and response time require calibration in the preparation.

Expanding σ about a field-free operating voltage v^* (mV) (a Volterra/Taylor expansion, as in [25]) gives $\sigma(v^* + s) \approx \sigma(v^*) + \sigma'(v^*)\,s + \frac{1}{2}\sigma''(v^*)\,s^2 + \ldots$ The truncation requires $|s| \ll 1/\rho \approx 1.8$ mV and a nonzero second derivative for quadratic mixing to be leading. Larger excursions require the full transfer curve. The linear term preserves the two input frequencies. The quadratic cross term generates their difference through the identity $2\cos\omega_1 t\cos\omega_2 t = \cos\Omega t + \cos(\omega_1+\omega_2)t$. Low-pass synaptic filtering suppresses the fast terms after mixing; membrane filtering, when retained, already shapes the carrier polarization. We isolate the slow source by a carrier average, before applying its frequency-dependent network gain. The fast $\cos^2\omega_j t$ terms average to $\frac{1}{2}$ and the sum-frequency term to zero:

$$\big\langle\sigma(v^* + s)\big\rangle_c = \sigma(v^*) + \tfrac{1}{4}\sigma''(v^*)\,(\varepsilon_1^2 + \varepsilon_2^2) + \tfrac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2\cos\Omega t + O(\varepsilon^4), \tag{5}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$Y_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon_1\varepsilon_2, \tag{6}$$

Here Y_Ω is the signed rate amplitude before recurrent feedback, in s^{-1}; ε denotes a common small amplitude scale, and $\simeq$ means to leading order in the polarizations, here and in what follows. The same expansion leaves a constant (DC) term, $\frac{1}{4}\sigma''(\varepsilon_1^2 + \varepsilon_2^2)$, which any read-out of the mean rate must include.

For a dynamic cellular response with synaptic input held fixed, define the effective second-order coefficient $\chi^{(2)}$ by the complex beat amplitude Y_Ω,

$$Y_\Omega \simeq \chi^{(2)}(\omega_2, -\omega_1)\,\varepsilon_2\varepsilon_1, \qquad r_\Omega(t) = \mathrm{Re}\Big[Y_\Omega e^{i\Omega t}\Big]. \tag{7}$$

Here $r_\Omega(t)$ is the generated rate component before recurrent feedback. Thus $\chi^{(2)}$ has units $\mathrm{s^{-1}mV^{-2}}$ when its input is polarization and its output is physical rate. With this cosine-amplitude convention, the memoryless limit is $\chi^{(2)} = \sigma''(v^*)/2$. Cellular processes contribute to this coefficient according to their state and response times. Weak input permits a second-order expansion, but a response that depends on input history retains frequency-dependent amplitude and phase. If symmetry or cancellation makes this coefficient zero, higher orders determine the leading beat.

Equation (6) carries the static detector's predictions: the generated beat is quadratic in the drive ($Y_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($Y_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \lambda_f E_0$, with applied amplitude E_0, so attenuating access raises the field required for a given effect, consistent with the carrier dependence of TI thresholds in [21, 22]. Carrier independence is thus a statement about the recovered beat component at fixed polarization, not about the applied field.

2.3 Envelope oscillations and spatial weighting

The same two-tone field gives different spatial dose maps under envelope following and quadratic detection. The distinction is the detector's response to the oscillating envelope, not a change in the applied field. Write $\mathbf{x}$ for position and $\varepsilon_j(\mathbf{x})$ for the signed local polarization of tone j, projected along a fixed direction. These amplitudes vary in space and are held fixed only during the temporal averages at one location. The nonnegative envelope is

$$\mathcal{A}(\mathbf{x}, t) = \sqrt{\varepsilon_1(\mathbf{x})^2 + \varepsilon_2(\mathbf{x})^2 + 2\varepsilon_1(\mathbf{x})\varepsilon_2(\mathbf{x}) \cos \Omega t}.$$

Carrier averaging of the squared signal gives

$$\mathrm{LPF}[s^2(t)] = \tfrac{1}{2}\mathcal{A}^2(t) = \frac{\varepsilon_1^2 + \varepsilon_2^2}{2} + \varepsilon_1\varepsilon_2 \cos \Omega t, \tag{8}$$

with spatial arguments suppressed at a fixed location. Thus square-law detection recovers $\mathcal{A}^2$, which is sinusoidal at Ω, whereas the envelope itself is not: at a matched site, $\mathcal{A}(t) = 2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine. The conventional TI map describes its excursion, $\mathcal{M}[y] \equiv \max_t y - \min_t y$, along the chosen direction [1]. Table 1 compares this envelope with the squared-envelope signal; the overbar denotes a beat-period mean. The geometric maps precede detector gains: an ideal envelope follower has response $c_1\mathcal{A}$ for a specified gain c_1. A smooth transfer g with nonzero curvature gives the weak response $g''(v^*)\mathcal{A}^2/4$; for the JR column, $g = \sigma$. The last row carries these gains into the signed steady source. In particular, the nonnegative mean-square polarization $\mathcal{I}$ alone does not specify excitation or suppression. Envelope following is an additional response assumption; the conventional envelope-excursion map alone does not specify a neuron's mean output.

The excursion maps in Table 1 are related pointwise by

$$\mathcal{M}[\tfrac{1}{2}\mathcal{A}^2] = \max(|\varepsilon_1|, |\varepsilon_2|)\, \mathcal{M}[\mathcal{A}]. \tag{9}$$

The square-law map is the conventional envelope-excursion map multiplied, at each point, by the dominant local polarization magnitude. It need not preserve the spatial contrast of the conventional TI map. Both maps can nevertheless have a central modulation maximum, as the illustrative line profile in Fig. S13a shows; the quadratic profile is broader in that example. Where the dominant carrier is larger superficially than at depth, this reweighting degrades deep-to-superficial contrast. As an illustrative example, at $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially against $(0.5, 0.5)$ at a matched deep target, $\mathcal{M}[\mathcal{A}]$ favors the superficial point by $2\times$ and $\mathcal{M}[\frac{1}{2}\mathcal{A}^2]$ by $40\times$. Focality must therefore be computed from the product map, whose spatial ranking can differ from the envelope map's. Its contrast depends on the actual vector field distribution: a sufficiently weak second carrier can

	Envelope following	Quadratic detection
Slow signal	$\mathcal{A}(\mathbf{x}, t)$	$\frac{1}{2}\mathcal{A}(\mathbf{x}, t)^2$
Mean before gain	$\overline{\mathcal{A}}(\mathbf{x})$	$\mathcal{I}(\mathbf{x}) = \dfrac{\varepsilon_1^2 + \varepsilon_2^2}{2}$
Modulation excursion	$2\min(\lvert\varepsilon_1\rvert, \lvert\varepsilon_2\rvert)$	$2\lvert\varepsilon_1\varepsilon_2\rvert$
Units before detector gain	mV	mV^2
Signed steady source (output units)	$c_1\overline{\mathcal{A}}$	$Y_{\mathrm{DC}} = \frac{1}{2}g''(v^*)\mathcal{I}$

Table 1: Envelope and quadratic maps at the same position $\mathbf{x}$. The first four rows give geometric quantities before detector gain; the last gives the steady source after gain, before recurrent feedback. Its sign depends on the detector. Averaging the slow signal over a beat period gives its mean. Excursion means maximum minus minimum, not Fourier amplitude.

suppress the superficial product. The line profile and two-site example illustrate the algebra; a realistic contrast requires fields computed on a head model.

With one fixed electrode pair per carrier and fixed neural state, changing the current ratio rescales the leading quadratic beat map without moving its maxima. For fixed frequencies, linear access, and fixed projection directions, write $\varepsilon_j(\mathbf{x}) = I_j^{\mathrm{pair}}\mathcal{P}_j(\mathbf{x})$, where $I_j^{\mathrm{pair}} > 0$ is the peak pair current (mA) and $\mathcal{P}_j$ is the signed local polarization per unit current (mV/mA). Then

$$\mathcal{M}[\tfrac{1}{2}\mathcal{A}^2] = 2I_1^{\mathrm{pair}} I_2^{\mathrm{pair}} |\mathcal{P}_1\mathcal{P}_2|.$$

The normalized map and its ratios between fixed target and reference regions are unchanged, as Cassarà et al. noted for second-order mixing [30]. Fixed detector coefficients and linear network propagation preserve this common current factor, including for dynamic mixing and spatially heterogeneous gains. Steering within this regime requires changing at least one carrier's field shape, for example by redistributing current among several electrodes at that frequency. The mean-square polarization map, however, weights $\mathcal{P}_1^2$ and $\mathcal{P}_2^2$ separately by the squared currents. It changes with the ratio; after signed detection, the resulting steady source can indirectly reshape the beat response by shifting local operating points. Threshold effects can also break the quadratic prediction; the conventional 2 min steering law applies when the detector follows the envelope. An interior beat maximum remains compatible with the fixed-state product law.

The network supplies dynamical selectivity in addition to this spatial weighting. Writing $R_\Omega(\mathbf{x})$ for the complex demodulated response at the envelope frequency Ω produced at location $\mathbf{x}$, and $G_{\mathbf{x}}(\Omega)$ for the local network susceptibility there,

$$R_\Omega(\mathbf{x}) \;\propto\; \sigma''(v_{\mathbf{x}}^*)\, G_{\mathbf{x}}(\Omega)\, \varepsilon_1(\mathbf{x})\, \varepsilon_2(\mathbf{x}), \tag{10}$$

a product of three factors. Field overlap sets where a difference-frequency drive is generated at all; the operating point and the network susceptibility set which of the driven populations amplifies it. Resonance does not override an unfavorable field product, but it can compensate for one.

Equation (10) is written for scalar projected fields. In general the coupling is vectorial and compartment-specific, $\varepsilon_i(\mathbf{x}) = \boldsymbol{\lambda}\cdot\mathbf{E}_i(\mathbf{x})$, so the mixing product $(\boldsymbol{\lambda}\cdot\mathbf{E}_1)(\boldsymbol{\lambda}\cdot\mathbf{E}_2)$ is signed: its magnitude follows the product of the projected amplitudes, while its phase inverts with either the sign of σ'' or the relative sign of the two projections. The scalar formulas use these fixed projections.

2.4 Detector laws and harmonic signatures

A neural response proportional to the squared envelope is not equivalent to one proportional to the envelope. A detector implements some transduction $T(\mathcal{A})$ of the local envelope $\mathcal{A}(t)$, and the

two laws predict different amplitude scaling, harmonic content and spatial weighting.

For a narrowband field $s(t) = \mathcal{A}(t)\cos\omega_c t$ passing through a memoryless nonlinearity g, the low-frequency (baseband) output Y_{LF} is the average of g over one carrier cycle, with carrier phase θ:

$$Y_{\mathrm{LF}}(\mathcal{A}) = (2\pi)^{-1}\int_0^{2\pi} g(v^* + \mathcal{A}\cos\theta)\,\mathrm{d}\theta.$$

Odd powers of $\mathcal{A}$ average away, leaving only even ones:

$$Y_{\mathrm{LF}}(\mathcal{A}) = g(v^*) + \tfrac{1}{4}g''(v^*)\mathcal{A}^2 + \tfrac{1}{64}g^{(4)}(v^*)\mathcal{A}^4 + \cdots \tag{11}$$

A smooth memoryless response with $g''(v^*) \neq 0$ therefore transduces $\mathcal{A}^2$ at leading weak-input order, whatever its detailed shape. For the JR column, $g = \sigma$, giving the mechanism of §2.2. Recovering $\mathcal{A}$ itself requires a qualitatively different detector: strong rectification. A full-wave rectifier has low-pass-filtered (LPF) output $\mathrm{LPF}[\,|\mathcal{A}\cos\omega_c t|\,] = (2/\pi)\mathcal{A}$ and a half-wave rectifier $(1/\pi)\mathcal{A}$, both linear in the envelope, as does a diode-and-capacitor peak detector. A hard threshold is not one of them: a step detector with offset $\delta > 0$ transduces $\pi^{-1}\arccos(\delta/\mathcal{A})$ for $\mathcal{A} > \delta$ and nothing below it, which is neither linear nor quadratic in $\mathcal{A}$. Asymmetric channel rectification, threshold crossing and release events are candidate biological realizations of rectifiers of one kind or another.

Envelope following and square-law detection are distinguishable at a single site. At a matched target, $\mathcal{A}(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos\theta| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1}\frac{(-1)^{n+1}}{4n^2-1}\cos(2n\theta)$ gives a harmonic ladder whose nonnegative Fourier amplitudes $A_{n\Omega}$ obey $A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}\mathcal{A}^2 = \varepsilon^2(1+\cos\Omega t)$ contains only DC and Ω. For the mixed family $T(\mathcal{A}) = c_1\mathcal{A} + c_2\mathcal{A}^2$, with $c_1 \neq 0$ and a nonzero fundamental, the pre-network ratio is

$$\mathcal{H}_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_\Omega}\right| = \frac{1/5}{\left|1 + \frac{3\pi}{4}\,c_2\varepsilon/c_1\right|}, \tag{12}$$

so measuring $\mathcal{H}_{\mathrm{det}}$ constrains $c_2\varepsilon/c_1$ within this family.[1]

A smooth detector generates a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, where $c_2 = g''(v^*)/4$ and $c_4 = g^{(4)}(v^*)/64$ are the coefficients in Eq. (11). The square-law signature is a harmonic ratio tending to zero with input strength. The recurrent network also multiplies each harmonic by its frequency-dependent gain G, $\mathcal{H}_{\mathrm{meas}} = \mathcal{H}_{\mathrm{det}}\,|G(2\Omega)/G(\Omega)|$. Near resonance, harmonic suppression can make an envelope follower appear square-law.

An amplitude sweep tests the scaling rather than one ratio. An envelope-following detector gives $Y_\Omega \propto \varepsilon$ with the harmonic ratio pinned at $1/5$; a weak square-law detector gives $Y_\Omega \propto \varepsilon^2$ with a ratio growing as ε^2. A threshold detector is a third class: its response vanishes for $2\varepsilon \leq \delta$, and its harmonic ratio varies with amplitude after onset. A single harmonic ratio cannot distinguish all three; amplitude dependence is the primary test.

2.5 The steady source has a state-dependent sign

Mean-square polarization is a nonnegative geometric map. Detection converts it into a signed steady source, and the circuit converts that source into a mean response. Under uniform conductivity, access, and detection, the source's largest magnitude lies on compartment boundaries; the envelope-following mean obeys the same restriction. The envelope mean is

$$\overline{\mathcal{A}}(\mathbf{x}) = \frac{1}{2\pi}\int_0^{2\pi}\sqrt{\varepsilon_1(\mathbf{x})^2 + \varepsilon_2(\mathbf{x})^2 + 2\varepsilon_1(\mathbf{x})\varepsilon_2(\mathbf{x})\cos\vartheta}\,\mathrm{d}\vartheta,$$

[1] The inverse has two branches unless the sign of the fundamental relative to the linear term is known; for $c_2/c_1 \geq 0$ it is unique.

where ϑ is beat phase. At a matched site with $\varepsilon_1 = \varepsilon_2 = \varepsilon > 0$, the envelope and quadratic means are $4\varepsilon/\pi$ and ε^2, respectively. The applied field has no DC component: these means describe the ideal detector outputs after multiplication by their gains, and the quadratic response change takes the sign of g''. Even one carrier gives either ideal detector a mean output, but no beat modulation; adaptation can alter the physiological mean response.

For the JR quadratic detector, the local rate source in Eq. (5) is

$$Y_{\mathrm{DC}}(\mathbf{x}, \mathbf{n}) = \frac{\sigma''(v^*_{\mathbf{x}})}{2}\, \mathcal{I}(\mathbf{x}, \mathbf{n}), \qquad \mathcal{I}(\mathbf{x}, \mathbf{n}) = \frac{\varepsilon_1(\mathbf{x}, \mathbf{n})^2 + \varepsilon_2(\mathbf{x}, \mathbf{n})^2}{2} \geq 0.$$

Here $v^*_{\mathbf{x}}$ is the baseline voltage at position $\mathbf{x}$ and $\mathbf{n}$ is the unit cell orientation used for both signed polarizations. The source Y_{DC} is positive on the convex part of the rate curve, negative on its concave part, and zero to this order at the inflection. Reversing either carrier's polarity or $\mathbf{n}$ leaves the squares unchanged; other directions change their magnitudes, while the sign remains set by the fixed curvature. Figure S13c shows both signs using one positive normalization. At the pinned JR operating point $v^* = 8.30$ mV, $\sigma'' < 0$, so the static source lowers the local rate. The observed mean-rate change also depends on which populations receive the source and on recurrent feedback.

Carrier averaging retains the cross term as the beat; averaging over many beat cycles removes it, leaving a sum of intensities without cross-frequency cancellation. At the detector output, where $Y_{\mathrm{DC}} \neq 0$, $|Y_\Omega / Y_{\mathrm{DC}}| = 2|\varepsilon_1\varepsilon_2|/(\varepsilon_1^2 + \varepsilon_2^2) \leq 1$, with equality at matched tone magnitudes. A common access factor cancels from this local ratio. In the polarization mean-square map $\mathcal{I} = (\varepsilon_1^2 + \varepsilon_2^2)/2$, the example above gives 50.5 superficially and 0.25 at depth, a 202-fold contrast; the beat/DC ratios are 0.20 and 1, respectively.

The geometric restriction follows in a homogeneous, isotropic, source-free conductor. Writing ϕ_j for the potential of tone j, the potentials ϕ_j, each component of $\mathbf{E}_j = -\nabla\phi_j$, and the signed polarizations $\varepsilon_j = \boldsymbol{\lambda} \cdot \mathbf{E}_j$ for spatially fixed coupling $\boldsymbol{\lambda}$ are harmonic. Thus $\nabla^2\mathcal{I} = \sum_j |\nabla\varepsilon_j|^2 \geq 0$: the intensity map is subharmonic and has no strict interior maximum. A montage can shape each carrier field, but the maximum of this map lies on the region's boundary; conductivity interfaces are such boundaries. For a spatially constant nonzero detector coefficient, this bounds the magnitude of Y_{DC}. A negative coefficient makes the source most negative at the boundary. Spatially varying coefficients require a separately weighted map.

The envelope mean has the same restriction. At each beat phase ϑ, $\mathcal{A}$ is the Euclidean norm of $(\varepsilon_1 + \varepsilon_2 \cos\vartheta,\, \varepsilon_2 \sin\vartheta)$, a vector of harmonic functions. A norm is convex, so this envelope is subharmonic; averaging over ϑ preserves that property. Thus both idealized mean maps lack a strict interior maximum under these conditions (Fig. S13b). This does not prohibit a beat maximum: the product instead obeys $\nabla^2(\varepsilon_1\varepsilon_2) = 2\nabla\varepsilon_1 \cdot \nabla\varepsilon_2$, whose sign is unrestricted. Within the homogeneous model, interior selectivity must therefore come from the oscillatory response rather than from either mean map.

Spatially varying access, orientation, or curvature can change the signed-source map and the resulting physiological mean-response map. The intensity result does not extend to these weighted responses. For isotropically distributed unit orientations $\mathbf{n}$ with orientation-independent access and curvature, $\langle(\mathbf{n} \cdot \mathbf{E}_j)^2\rangle = |\mathbf{E}_j|^2/3$ and $\langle(\mathbf{n} \cdot \mathbf{E}_1)(\mathbf{n} \cdot \mathbf{E}_2)\rangle = \mathbf{E}_1 \cdot \mathbf{E}_2/3$: orthogonal fields cancel the averaged beat, but retain the DC intensity. At equal local RMS polarization, an unmodulated carrier gives the same mean shift at second order when its DC mixing coefficient equals those of both TI carriers. For a dynamic detector, this requires the frequency-dependent coefficient to vary negligibly across the carrier interval.

2.6 Amplification: the synaptic low-pass and the network resonance

Synaptic filtering suppresses the carrier after the nonlinearity has generated the beat. Each JR excitatory synapse has transfer function $H_s(\omega) = Aa/(i\omega + a)^2$, a second-order low-pass with

characteristic frequency $a/(2\pi) \approx 16$ Hz. It passes a 10 Hz component with relative amplitude about 0.7, while reducing a 100 Hz component to about 2.5% (Fig. 6). Provided the carrier does not itself excite a responsive mode, the loop therefore reads predominantly the generated low-frequency drive.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the JR column (node) about its fixed point gives a leading eigenvalue pair $\mu_\pm = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, with damping γ and modal frequency ω_0. Below a supercritical Hopf, $\gamma \to 0^+$ and the column is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (6) at Ω gives a resonant response. Writing G for the complex susceptibility, a constant-numerator two-pole approximation gives its magnitude as

$$|G(\Omega)| \propto \left[(\omega_0^2 + \gamma^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad |G(\omega_0)| \propto \frac{1}{2\gamma\omega_0} \quad (\gamma \ll \omega_0), \tag{13}$$

so the response peaks near ω_0 with an approximately Lorentzian power profile and $1/\gamma$ growth near a simple Hopf with nonzero input–output projections.[2] This weak-drive law requires the rectified mean shift to leave the state stable; above the Hopf, entrainment means capture of an autonomous rhythm and requires tracking its phase relative to the drive.

Write D for the detector's beat-source coefficient: for the JR rate source, $D = \frac{1}{2}\sigma''(v^*)$, half the local sigmoid curvature. Combining access, detection, and amplification gives

$$R_\Omega \simeq G(\Omega;\gamma)\; D\; \lambda_f(\omega_1)E_1\, \lambda_f(\omega_2)E_2, \tag{14}$$

where λ_f maps field to local carrier polarization, D converts the product of those polarizations into a beat drive, and G maps that drive to the complex observed response, whose measured amplitude is $A_\Omega = |R_\Omega|$. For JR, G is the response to a generated rate drive. A static detector has carrier-independent D at fixed polarization; a dynamic detector can retain frequency dependence. A carrier sweep and a beat sweep therefore constrain different stages rather than a single effective gain.

Resonance requires a beat source. The cubic Hopf normal form responds to an in-band drive but has no second-order difference-frequency source. A large resonant susceptibility alone therefore does not establish beat generation.

2.7 Carrier access depends on the effective membrane response

Carrier independence at fixed polarization does not imply independence at fixed applied field. As an isopotential access model, consider $H_m(\omega) = 1/(1 + i\omega\tau_m)$, with effective time constant τ_m. This is the response of a point membrane to injected current; a field coupled through a cable need not have the same transfer. For nearby carriers, the two polarizations acquire nearly the same attenuation. The quadratic beat then satisfies

$$\frac{|Y_\Omega(f_c)|}{|Y_\Omega^{\text{direct}}|} \simeq |H_m(\omega_c)|^2 = \frac{1}{1 + (\omega_c\tau_m)^2} \sim \frac{1}{(\omega_c\tau_m)^2} \quad (\omega_c\tau_m \gg 1), \tag{15}$$

where Y_Ω^{direct} is the beat generated with the same static coupling and no access attenuation. This filter belongs either in the field-to-polarization map or in the membrane equation that computes that polarization.

Somatic passive time constants are commonly 10–30 ms; reported human pyramidal values include approximately 12 ms [4] and 16.5 ± 3.7 ms [5]. For the illustrative $\tau_m = 16$ ms used here, the power

[2] Here ω_0 is the eigenvalue's imaginary part. The canonical oscillator has undamped frequency $\sqrt{\omega_0^2 + \gamma^2}$ and peak $\sqrt{\omega_0^2 - \gamma^2}$ for $\omega_0 > \gamma$; frequency-dependent network projections can shift the peak.

factor $|H_m|^2$ is about 10^{-4} at 1 kHz, 2.5×10^{-5} at 2 kHz, and 10^{-6} at 10 kHz. A static nonlinearity placed after this point-membrane response therefore receives strongly attenuated carriers.

Cohen et al.'s periaxonal circuit provides a reason to distinguish axonal from somatic access [7]. A conducting submyelin space, about 12 nm wide and partly sealed at the paranodes, separates the voltage across myelin from that across the axonal membrane. Their minimum fitted axonal decay was approximately 1 ms. This is a response to injected current in a distributed cable, rather than a measured transfer from a TI field. The AIS is another candidate site, with a high sodium-channel density [8]; this property supports nonlinear sensitivity but does not fix its charging time.

A shorter effective time changes carrier transmission quantitatively. The sensitivity case $\tau_m = 0.2$ ms gives $|H_m|^2 \approx 0.39$ at 1 kHz and 0.14 at 2 kHz (Fig. S5). This assumed time represents the fast-compartment hypothesis in Eq. (4), or motivates a shorter membrane time in a QIF reduction; it has not been measured under TI at the AIS or a node. In the latter description, the approach to a static transfer is governed by Eq. (19); the membrane filter must not be applied again outside the QIF equations. A short access time improves carrier transmission without, by itself, specifying the mixing coefficient or the remaining circuit resonance. Table 2 separates measured decays, assumed access times, strong-pulse charging, cellular detection, and downstream synaptic filtering. A characteristic frequency requires a specified filter; a measured decay alone is not a TI cutoff.

Response	Time scale	Evidence and role
Somatic point-membrane model	$\tau_m = 16$ ms; $f_{\text{char}} \approx 10$ Hz	Illustrative current-driven filter, informed by somatic measurements near 12 and 16.5 ms [4, 5].
Myelinated axon	minimum fitted decay ≈ 1 ms	Injected-current responses in Cohen et al.'s periaxonal circuit [7]; evidence for distinct axonal voltage dynamics.
Fast axonal access	assumed $\tau_m = 0.2$ ms; $f_{\text{char}} \approx 800$ Hz	Reduced-model sensitivity case for a fast compartment, such as an AIS or node; TI coupling and charging time require calibration.
Prior TI network model	$\tau_m = 1$ and 0.4 ms	Short effective membrane times governing kHz sensitivity in a gamma network [6]; model parameters, not measured axonal TI decays.
Field-induced charging	≈ 0.7 μs	Measured with strong microsecond field pulses in isolated neurons [23]; a different stimulation regime.
Cellular detector	channel- and state-dependent	Luff et al.'s TTX-sensitive mixing persists near 5 kHz [10]; this measures detection, not a passive time constant.
JR excitatory synapse	$\tau_s = 10$ ms; $f_{\text{char}} \approx 16$ Hz	Model filter acting on the generated rate, downstream of detection.

Table 2: Access, detection, and downstream filtering involve different response times. Measured decays, pulse-charging times, and assumed model parameters are identified separately. The characteristic frequency $f_{\text{char}} = 1/(2\pi\tau)$ is given only for the specified filters; an axonal decay is not automatically a TI cutoff. AIS, axon initial segment; TTX, tetrodotoxin.

Cable geometry changes this attenuation: a long passive cable with a sealed end has a quadratic-beat factor of about 10^{-2} at 1 kHz and 10^{-3} at 10 kHz for $\tau_m = 16$ ms, rather than the point-membrane values for a sufficiently long, sealed cable in a uniform axial field.

Channel kinetics contribute to the effective dynamic mixing response. Any separate gate filter needs a model or measurement and must represent kinetics absent from the detector. QIF, for example, lacks explicit sodium-channel kinetics. Neither QIF nor a static sigmoid specifies a universal 1 kHz cutoff. Carrier sweeps can constrain the combined access and mixing response,

while beat-frequency sweeps probe the downstream amplifier.

2.8 Operating-point dependence

Because $Y_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 7). The Population Excitation–Inhibition Index (PEIX) of [24] measures signed departure from maximal sigmoid slope:

$$\mathrm{PEIX}(v^*) = \mathrm{sign}(v^* - v_0)\left\{1 - \mathrm{sech}^2\left[\tfrac{\rho}{2}\,(v^* - v_0)\right]\right\}. \tag{16}$$

It is zero at v_0, positive above it, negative below it, and tends to ± 1 in the saturated tails. The normalized slope is instead $\sigma'(v^*)/\sigma'(v_0) = 1 - |\mathrm{PEIX}(v^*)|$, with $\sigma'(v_0) = e_0\rho/2$. Demodulation follows the curvature, which vanishes at the inflection and in the tails and peaks in magnitude on the flanks. For the logistic curve, $\sigma'' = -\rho\sigma' \,\mathrm{sign}(\mathrm{PEIX})\sqrt{|\mathrm{PEIX}|}$; the index specifies the side and saturation, but is not itself the mixing gain.

PEIX is a measurable operating-point index; its relation to mixing is specific to the memoryless logistic detector. In the Jansen–Rit detector, anything shifting v^* (for example, neuromodulation, attention, arousal, psychedelic exposure, or PV-interneuron dysfunction in Alzheimer's disease) scales, and can flip the sign of, the TI response; a dynamic detector responds through its own state-dependent coefficient and susceptibility, without a universal sign rule. State changes therefore test detection only when changes in recurrent gain are measured as well.

2.9 The exact mean field extends the static detector

The static rate curve applies when population activity follows the input fast enough. An exact mean field of quadratic integrate-and-fire (QIF) neurons makes this condition explicit and supplies the response outside that limit.

The corresponding population nonlinearity can be derived for all-to-all coupled QIF neurons with Lorentzian-distributed excitabilities. Their exact Montbrió–Pazó–Roxin (MPR) mean field evolves the firing rate r and mean voltage v [31]; adding dynamic synapses gives a second-generation neural mass model (NMM2) [32, 33]. With membrane time τ_m, physical firing rate r, and dimensionless voltage v,

$$\tau_m \dot{r} = \frac{\Delta}{\pi\tau_m} + 2rv, \qquad \tau_m \dot{v} = v^2 + \bar{\eta} - (\pi\tau_m r)^2 + I(t), \tag{17}$$

where $\bar{\eta}$ is the mean excitability, $\Delta > 0$ its half-width, and $I(t)$ the dimensionless input, including recurrent drive when present. Dots denote physical-time derivatives. For a constant total drive $Q = \bar{\eta} + I$, setting both derivatives to zero gives $v = -\Delta/(2\pi\tau_m r)$ and

$$\tau_m r = F(Q) = \frac{1}{\pi}\sqrt{\frac{Q + \sqrt{Q^2 + \Delta^2}}{2}}. \tag{18}$$

Primes on F denote derivatives with respect to the total drive Q. Below its inflection, curvature reflects recruitment of cells across threshold; above it, curvature reflects the concave square-root rate of active QIF neurons. This nonlinear curve exists without recurrent coupling. It has an inflection at $Q = \Delta/\sqrt{3}$ and an unbounded concave rise, unlike the saturating JR sigmoid.

If the population relaxes to this equilibrium faster than synaptic inputs change, NMM2 reduces to a static rate map with dynamic synapses [32, 33]. A second-order synapse with time constant τ_s obeys

$$r(t) \simeq \frac{F(Q(t))}{\tau_m}, \qquad \tau_s^2 \ddot{r}_{\mathrm{syn}} + 2\tau_s \dot{r}_{\mathrm{syn}} + r_{\mathrm{syn}} = r(t), \tag{19}$$

where r_{syn} is the filtered rate entering the recurrent drive in Q, with the original coupling normalization retained. This is the slow-synapse limit: shortening the effective membrane time makes the

transfer appear static while the network retains its synaptic dynamics. The relevant separation is between the population relaxation time $\tau_{\rm rel}$ and τ_s. Applying the same static map directly to a carrier of angular frequency ω_c requires the stronger condition $\omega_c\tau_{\rm rel} \ll 1$. A population can therefore be quasistatic for its synaptic inputs yet dynamic for a kHz carrier.

Recurrent excitatory–inhibitory circuits can resonate without intrinsic cellular resonance [13, 14]. In the fast-membrane limit, synaptic modes remain, whereas resonances requiring finite membrane dynamics need not survive.

Outside the adiabatic limit, the quadratic voltage term mixes the carriers before the rate responds. For an uncoupled population under a weak current $I_{\rm ac}\cos\omega t$, write the mean-rate shift as $I_{\rm ac}^2\kappa_{\rm DC}/\tau_m$. At slow carriers, $\kappa_{\rm DC} \to F''(Q)/4$; at fast carriers, $\kappa_{\rm DC} \sim F'(Q)/[2(\omega\tau_m)^2]$. The slow response follows curvature and can be negative above the inflection. The fast DC coefficient instead follows F', which is largest at the inflection; it has no PEIX-based sign reversal. The fast response follows the positive slope: voltage excursions of order $I_{\rm ac}/(\omega\tau_m)$ generate a positive input shift through v^2. A shorter membrane time can therefore change both the magnitude and sign of the mean response at a given physical carrier frequency.

The fast-carrier attenuation already includes membrane access. Dividing by the squared voltage excursion leaves a coefficient tending to $F'/2$. For two nearby tones, the voltage quadratic generates a beat source proportional to $I_1I_2/(\omega_1\omega_2\tau_m^2)$, where I_j is the current amplitude of tone j, with the same product law as the static detector. The dynamic rate equation attenuates the carriers once more, after this mixing; it does not add that attenuation to the generated beat. The exact coefficients and carrier exponents are derived in the supplement.

In membrane-time units, the fast-carrier QIF voltage source has $D = 1$; G then includes its conversion through the voltage and rate dynamics. QIF thus provides a cellular realization of quadratic detection, with its own frequency dependence. Other smooth cellular mechanisms can contribute to the same effective second-order coefficient. Their weak-input product law does not fix the coefficient's magnitude, phase, or carrier dependence.

3 Methods

3.1 Stimulation waveforms and matched-source controls

Dense perturbative sweeps use the three-line waveform

$$s(t) = \varepsilon\,[\,1 + m\cos\Omega t\,]\cos\omega_c t, \qquad \omega_c = 2\pi f_c, \tag{20}$$

with dimensionless modulation depth m, because it parameterizes a sweep by one amplitude. It is not the physical signal: it carries a line at ω_c that the two-tone sum lacks. The two agree at Ω to leading order (the surrogate gives $Y_\Omega = \frac{1}{2}\sigma''\varepsilon^2 m$) and are *not* equivalent beyond it. For matched genuine tones $\mathrm{LPF}[s^2] = \varepsilon^2(1+\cos\Omega t)$, whereas for the $m = 1$ AM surrogate $\mathrm{LPF}[s^2] = \varepsilon^2(\frac{3}{4} + \cos\Omega t + \frac{1}{4}\cos 2\Omega t)$: the same Ω amplitude, but a different DC term and a spurious second harmonic at 25% of the fundamental. Genuine two tones are used for the demodulation demonstration (Fig. 4), the curvature and square-law controls (Fig. 7), the laminar maps (Figs. S1, S7) and the kHz demonstration. The AM surrogate is retained for dense resonance and coupling sweeps; Fig. 7d verifies open-loop agreement at Ω at one operating point. That agreement does not extend to the mean shift or second harmonic, the latter being a diagnostic of transduction class (§2.4). We therefore repeat the mean-shift-dependent comparisons with two tones: the timing-versus-rate figure uses them, as do controls of the TI/tACS ratios and NMM2 endpoint. The corresponding Results report both waveforms, whose different DC terms can shift the recurrent operating state.

3.2 Amplitude scaling and harmonic measurements

These measurements distinguish quadratic detection from envelope following, which predict different dose maps and cannot be identified from a beat response at one amplitude. With fixed coupling and a linear downstream response, an amplitude sweep tests whether the generated beat scales as ε^2 or ε. The harmonic test compares the response at the second harmonic, 2Ω, with the fundamental at Ω. After correcting for network gain, their ratio tends to zero for weak quadratic detection but to 1/5 for ideal envelope following. The amplitude law and harmonic ratio jointly identify the detector class.

The response at the envelope frequency is read out with a software lock-in amplifier: multiplying $v(t)$ by sine and cosine at Ω and averaging estimates the component at that frequency. A finite window suppresses neighboring frequencies without eliminating them. With a Hann window, the in-phase ($L_{\cos}$) and quadrature ($L_{\sin}$) components are

$$L_{\cos} = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \cos \Omega t_k, \quad L_{\sin} = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \sin \Omega t_k,$$
$$A_\Omega = \sqrt{L_{\cos}^2 + L_{\sin}^2}, \tag{21}$$

with sample index k, sample times t_k, voltages $v_k = v(t_k)$, and Hann weights w_k. The windowed mean is $\bar{v} = \sum_k w_k v_k / \sum_k w_k$. For a test tone $v = A_{\text{test}} \cos(\Omega t + \theta_0)$, with amplitude A_{test} and phase θ_0, the long-window limits $(L_{\cos}, L_{\sin}) \to A_{\text{test}}(\cos\theta_0, -\sin\theta_0)$ give $A_\Omega \to A_{\text{test}}$: the normalization returns the physical amplitude, twice the magnitude of the windowed Fourier coefficient. Out-of-band components can leak into the estimate, particularly the DC level of the JR voltage readout ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). The finite-window projection of a constant onto the quadrature references need not vanish, so we subtract $\bar{v}$ explicitly; this makes the metric accurate down to the 10^{-8} level needed for the JR kHz analysis.

The log–log amplitude slope distinguishes linear from quadratic scaling without absolute calibration.

To measure that harmonic ratio, apply Eq. (21) at Ω and 2Ω over the same settled window. Their amplitude ratio estimates $\mathcal{H}_{\text{meas}}$; correcting for $|G(2\Omega)/G(\Omega)|$ gives $\mathcal{H}_{\text{det}}$. Inferring a mixture coefficient through Eq. (12) additionally requires the mixed family $T(\mathcal{A}) = c_1\mathcal{A} + c_2\mathcal{A}^2$ and a known sign branch. A threshold response can give a similar ratio at one amplitude, so the amplitude sweep must also be checked.

3.3 Head model and field computation

To compare the spatial predictions on the same anatomy, we compute all maps in one open-access subject, IXI002 of the IXI dataset[3] (T1-weighted MRI; female, 35 years). The T1 image was segmented into six tissues (scalp, skull, cerebrospinal fluid, gray matter, white matter and eyes) and meshed with SimNIBS 3.2.3 `headreco` [34], with its CAT12 option [35]. Tissue conductivities were 0.33 (scalp), 0.008 (skull), 1.79 (cerebrospinal fluid), 0.40 (gray matter), 0.15 (white matter) and 0.33 S/m (eyes) [36]. These are low-frequency values; no dispersion correction was applied at the kHz carriers. The volume conductor is linear and quasi-static, so the carrier frequencies do not enter the maps.

The electrodes follow the published hippocampal montage of Violante et al. [37], placed by its recipe on the subject's registered 10–10 positions. Pair 1 (anterior) joins e1, 25 mm from T9 toward FT9, to e2, 10 mm from AF8 toward Fp2. Pair 2 (posterior) joins e3, 25 mm from T9 toward TP9, to e4, 10 mm from P8 toward PO8. Each electrode is a 20-mm disc of 3 mm of gel (4.0 S/m) under 2 mm of metal (10 S/m), meshed into the head model with SimNIBS 3.2.3; the resulting mesh has 3.64×10^6 linear tetrahedra. As a montage sensitivity, the 10–10 rendering

[3] `https://brain-development.org/ixi-dataset/`

of this montage, FT7–F8 (anterior) and TP7–P8 (posterior), which Yatsuda et al. [38] used as their starting montage, was meshed and solved in the same way. Each montage has its own mesh, carrying only its own four electrodes.

Each pair was solved once at 1 A. The potential ϕ_j (V) satisfies $\nabla\cdot(\sigma_{\text{cond}}\nabla\phi_j) = 0$, where σ_{cond} is tissue conductivity (S/m). Current enters and leaves through each electrode's metal contact as a Neumann condition, distributed in proportion to nodal area; the rest of the scalp carries no normal current, and one pinned node fixes the gauge. The linear finite-element system was assembled and solved with Neuroelectrics' temporal-interference library, by conjugate gradients with algebraic-multigrid preconditioning, to a relative residual of 10^{-8}. The field $-\nabla\phi_j$ was recovered at gray- and white-matter nodes by superconvergent patch recovery [39], separately within each compartment (446,877 node–compartment pairs). Dividing these fields by the solve current gives the field-per-current vectors $\mathbf{F}_j$, expressed in (V/m)/mA to match I_j^{pair}. Every current condition is a scaling of these two fields. With peak pair currents I_1^{pair} (anterior) and I_2^{pair} (posterior) summing to 4 mA, the ratio 1:1 is 2 + 2 mA and 1:3 is 1 mA anterior plus 3 mA posterior, the two conditions of Violante et al. The ratio sweep sets the posterior share $I_2^{\text{pair}}/(I_1^{\text{pair}} + I_2^{\text{pair}})$ from 0.1 to 0.9 in steps of 0.025.

The field-to-polarization conversion follows §2.3: $\varepsilon_j(\mathbf{x}) = I_j^{\text{pair}}\mathcal{P}_j(\mathbf{x})$, with $\mathcal{P}_j = \lambda_0\,\mathbf{F}_j\cdot\mathbf{n}$, one uniform static coupling $\lambda_0 = 0.5$ mV per V/m [40], and $\lambda_f \approx \lambda_0$. A uniform coupling multiplies the envelope map by λ_0 and the two second-order maps by λ_0^2. Target-to-reference ratios, normalized maps and focus positions do not depend on it. Hippocampal couplings of 0.12–0.18 mV per V/m [41, 42] would scale the absolute values by 0.24–0.36 and 0.06–0.13, respectively.

The detector compartment fixes the projection direction. A laminar direction for a pyramidal axon initial segment, a local axon direction for a node, or an orientation distribution for terminals defines a different access model. Isotropic averaging is one possible distribution, not an anatomical consequence of choosing terminals. Maximizing a geometric map over orientation gives an upper bound for that map at each location; it does not identify the detector's orientation.

The projection direction $\mathbf{n}(\mathbf{x}) = \mathbf{n}^*(\mathbf{x})$ is the envelope-maximizing direction at the reference ratio 1:1: the closed-form maximizer, over unit vectors, of $2\min(|\mathbf{F}_1\cdot\mathbf{n}|, |\mathbf{F}_2\cdot\mathbf{n}|)$ [1]. It is chosen once at each node and then held fixed for 1:1, 1:3 and every ratio of the sweep, and all three maps use it. Along it, the 1:1 envelope map equals its maximum over orientation. At any other ratio it is the envelope seen by an element with that fixed orientation, which can be below that ratio's own maximum. As an orientation sensitivity, every statistic was repeated along a second fixed field, the radial direction from the brain centroid. The isotropic comparison averages the signed mixing term over uniformly distributed orientations before its magnitude is taken: $\langle\varepsilon_1\varepsilon_2\rangle_{\mathbf{n}} = \lambda_0^2 I_1^{\text{pair}} I_2^{\text{pair}}\,\mathbf{F}_1\cdot\mathbf{F}_2/3$, so the isotropic beat excursion is $\frac{2}{3}\lambda_0^2 I_1^{\text{pair}} I_2^{\text{pair}}|\mathbf{F}_1\cdot\mathbf{F}_2|$. Elements whose beats have opposite phase cancel in this average.

The target is the left hippocampus of a FreeSurfer 6.0.0 `recon-all` segmentation of the same T1 image (`aseg` label 17, 3903 mm^3) [43]. The region of interest (ROI) is the 1,568 gray- and white-matter nodes inside it. The reference is the 273,544 gray-matter nodes outside the ROI, including other deep gray matter. It serves as a cortical-reference proxy. The mesh and the segmentation share one image frame; tissue-volume, centroid and white-matter checks confirmed their alignment. As a label sensitivity, the Neuromorphometrics atlas hippocampus, warped into the subject with SimNIBS 3.2.3 `mni2subject` and the subject's nonlinear MNI transform (1,345 nodes; Dice 0.72 with the FreeSurfer label), replaced the ROI. As a tissue sensitivity, the ROI was restricted to its gray-matter nodes (1,174 with the FreeSurfer label, 1,013 with the atlas label).

The target-to-reference ratio is the ROI 95th percentile divided by the reference 95th percentile; its mean-based companion divides the ROI mean by the same reference percentile. All percentiles and centroids weight nodes equally. The quadratic excursion and $\mathcal{I}$ are second order in the field, whereas the envelope excursion is first order. To separate this difference in degree from the spatial

weighting, each second-order statistic is also given degree-matched, computed on the square root of the map. This is a geometric comparison; the raw maps remain the respective detector predictions before gain. The focus is the centroid of the top 10% of ROI nodes, ranked by the map's own value, with 5, 25 and 50% as a sensitivity. Ranking makes the focus exactly invariant to a uniform positive factor. Its displacement from 1:1 is measured along the hippocampal long axis, the first principal axis of the ROI nodes, with anterior positive. Field projections still use $\mathbf{n}^*$, independently of this reporting axis. The 1:1 to 1:3 displacement was computed for every combination of montage, label, ROI tissue definition and focus fraction (32 per map), each montage along its own 1:1 direction.

No signed DC-source or mean-response map is computed. For a spatially uniform detector coefficient, $Y_{\mathrm{DC}} = \frac{1}{2}\sigma''(v^*)\mathcal{I}$ (§2.5) is $\mathcal{I}$ rescaled, with the sign of $\sigma''(v^*)$. A nonuniform coefficient would require an operating-point map, which a head model does not provide, and a mean response would additionally require a circuit model. In particular, $\sigma''(v^*)$ is curvature at the population model's operating point, not a measurement of the transfer at the axon initial segment. Assigning that coefficient to an anatomical detector requires a calibrated access and input–output model.

3.4 Single-column Jansen–Rit: integration, lock-in, and controls

We integrate the JR column as six first-order ordinary differential equations. The TI field enters only the pyramidal output sigmoid, Eq. (2), as $s(t)$, while the readout $v = y_1 - y_2$ contains only synaptic potentials. A beat in this readout must therefore have passed through the nonlinearity and the recurrent loop. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta (RK4) scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S9 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and 1.5×10^{-5} s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.7 are analytic evaluations of Eq. (15), not simulations.

The resonance sweep in Fig. 5 uses 10 s of settling and a 20 s measurement window. Long windows reduce transients from weakly damped modes and average beating when autonomous activity persists. Other runs use the windows specified in their scripts. The two-tone 2 kHz example (Fig. S6) uses $p = 400$ s^{-1}, $C = 135$, and 0.5 mV per tone with $\tau = 0.2$ ms. From a zero initial state, it settles for 4 s and records 4 s at $\Delta t = 1.5 \times 10^{-5}$ s, sampling before each RK4 step. Its amplitude spectra use plain-mean subtraction and a Hann window. Parameter sweeps advance all N_{grid} combinations simultaneously in a vectorized $(N_{\mathrm{grid}}, 6)$ state array.

The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically and diagonalizing it: the Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\mu_+$ in the alpha band. This gives $p_H \approx 315.70$ s^{-1} and $f_0 = \mathrm{Im}\,\mu_+/2\pi \approx 11.16$ Hz, consistent with the continuation and the JR analysis of [44]. The 11.1 Hz demonstration beat is near this resonance. For $p > p_H$ the fixed point is a stable focus.

We compare the *open-loop detector*, the firing-rate sigmoid at a fixed operating point v^* without recurrence, with the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the two-tone field through σ at v^* and applies Eq. (21), isolating the pure demodulation law (Figs. 6, 7, S5); its signed in-phase component $L_{\cos}$ verifies the curvature law, including its sign. The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in the pyramidal sigmoid in Eq. (2) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against

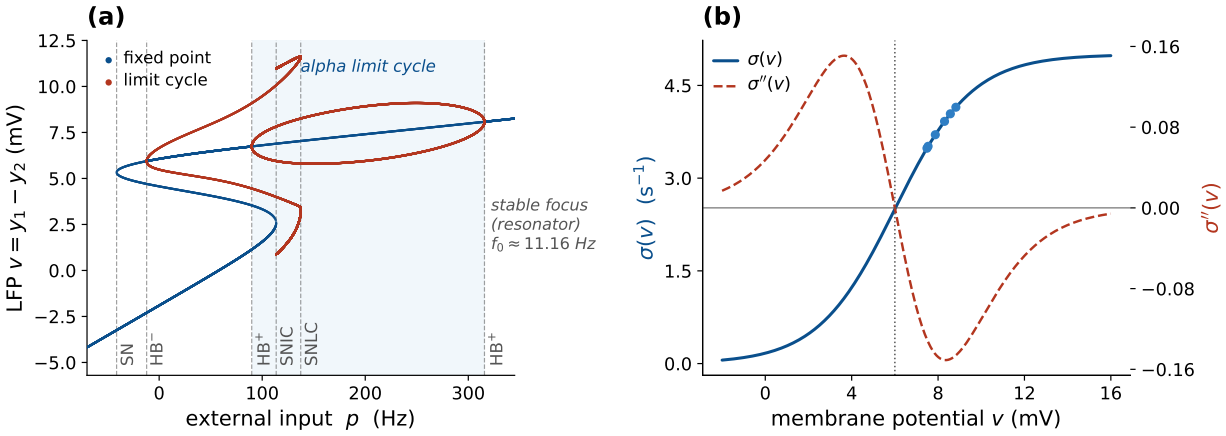


Figure 2: JR operating states and sigmoid curvature. *(a)* Numerical continuation of fixed points (blue) and cycle extrema (red) against external drive p; solid branches are stable, faded branches unstable. The alpha cycle lies between Hopf points near 90 and 316 s^{-1}. Beyond the upper Hopf the fixed point is a stable focus, with $f_0 \approx 11.16$ Hz. Markers identify saddle-nodes (SN), Hopf points (HB$^{\pm}$), the saddle-node on an invariant circle (SNIC), and the saddle-node of limit cycles (SNLC). *(b)* Sigmoid and curvature; dots mark the modeled operating points.

finite differences. Code and reproduction limits are listed under *Code and data availability.*

3.5 Exact next-generation mean field (NMM2 PING)

The pyramidal–interneuron network gamma (PING) circuit couples excitatory and inhibitory MPR populations through second-order synapses [31, 32, 33]. We use the Rosetta Stone formulation [26], which places this derived mean field on the same footing as the JR and laminar models. For population $X \in \{E, I\}$ (excitatory or inhibitory), r_X is the physical firing rate (ms^{-1} in the integration), v_X the dimensionless mean QIF voltage, and τ_X the membrane time (ms). Dots denote physical-time derivatives:

$$\begin{aligned} \tau_E \dot{r}_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, & \tau_E \dot{v}_E &= v_E^2 + \bar{\eta} - (\pi\tau_E r_E)^2 + \tau_E J\,(r_E^{\mathrm{syn}} - r_I^{\mathrm{syn}}) + I_E(t), \\ \tau_I \dot{r}_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, & \tau_I \dot{v}_I &= v_I^2 + \bar{\eta} - (\pi\tau_I r_I)^2 + \tau_I J\,(r_E^{\mathrm{syn}} - 2 r_I^{\mathrm{syn}}), \end{aligned} \tag{22}$$

with second-order synaptic activations $r_E^{\mathrm{syn}} = \hat{K}_a[r_E]$ (excitatory, time constant τ_a) and $r_I^{\mathrm{syn}} = \hat{K}_g[r_I]$ (inhibitory, τ_g), each obeying the same critically damped operator as the JR synapse, $\tau_s^2 \ddot{r}_{\mathrm{syn}} + 2\tau_s \dot{r}_{\mathrm{syn}} + r_{\mathrm{syn}} = r$; here J is the dimensionless global synaptic gain and the relative weights $(w_{EE}, w_{EI}, w_{IE}, w_{II}) = (1, 1, 1, 2)$ use target X and source Y in w_{XY} and set the E and I rows: the E membrane sees $J\,(r_E^{\mathrm{syn}} - r_I^{\mathrm{syn}})$ and the I membrane $J\,(r_E^{\mathrm{syn}} - 2r_I^{\mathrm{syn}})$, the only asymmetry being the doubled $I \to I$ weight $w_{II} = 2$.

In the schematic, $\mathcal{U}_j(t)$ is the total synaptic and field input to QIF neuron j, and J^* denotes a coupling-induced Hopf. The common background drive $\bar{\eta}$ is the bifurcation parameter: raising it past a supercritical Hopf at $\bar{\eta}_H = 1.110$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory voltage equation as a current-like forcing, $I_E(t) = \lambda_I E(t)$ in the convention of the exact mean field [32], with instantaneous field-to-current coupling λ_I, as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) of current amplitude A_f in the dimensionless MPR drive units, the same normalization as $\bar{\eta}$. Thus λ_I is measured in normalized drive per V/m. The E membrane $\tau_E \dot{v}_E$ integrates this forcing into the carrier-frequency excursion $A_f/(\omega\tau_E)$ that the exact quadratic v_E^2 term rectifies, with no postulated sigmoid; the PING as run therefore computes membrane access with $\tau_m = \tau_E$.

We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (21)) of r_E at Δf

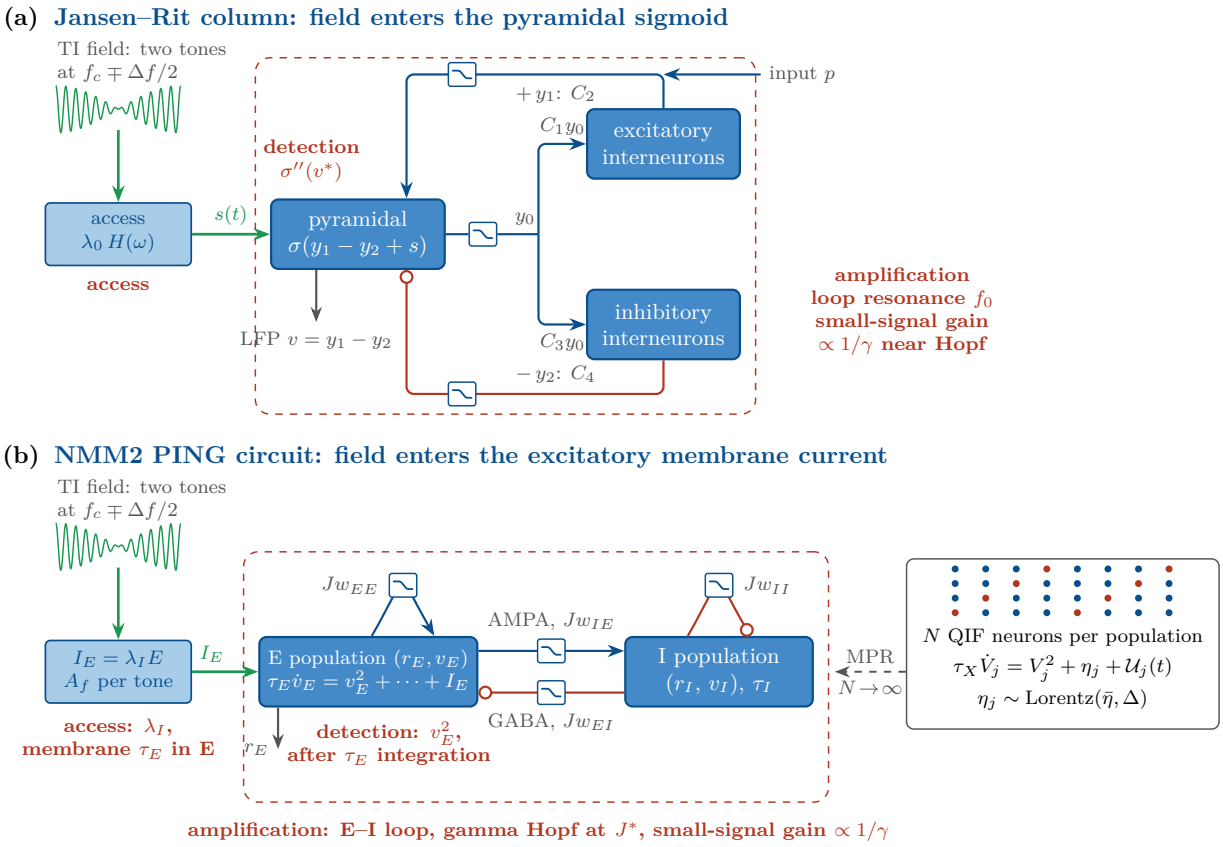


Figure 3: Access, detection, and amplification in the two circuits. *(a)* JR: access $\lambda_0 H(\omega)$ converts the field to polarization s, and sigmoid curvature mixes the tones. *(b)* NMM2 PING: current $I_E = \lambda_I E$ enters the excitatory membrane; v_E^2 mixes its voltage excursions. The inset shows the QIF neurons whose infinite-population limit gives this exact mean field. Low-pass symbols mark second-order synapses (h_e, h_i in a; AMPA and GABA with times τ_a, τ_g in b). They suppress carriers more strongly than the generated beat and transmit the steady shift. Dashed frames enclose recurrent amplifiers, with small-signal gain proportional to $1/\gamma$ near the relevant Hopf. Blue: excitation; red: inhibition; green: field input.

over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [26]): $\Delta = 1$, global coupling $J = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz), located by bisection on the leading eigenvalue of the mean-field Jacobian. The simulations retain these finite membrane times; Eq. (19) characterizes the fast-membrane limit.

As in JR, either background drive or coupling can bring the circuit to its Hopf. The resonance map sweeps $\bar{\eta}$ (Fig. S8); the coupling sweep, called the J-curve, varies J at fixed $\bar{\eta}$ toward J^*, located by bisection on the leading Jacobian eigenvalue, and reads the lock-in peak of r_E over Δf (Fig. S10). Unlike the JR J-curve, this sweep does not pin the operating point. At $\bar{\eta} = 0$, we track the eigenpair that reaches the Hopf by continuation and define γ from that critical pair, even where another mode dominates the response. The linear reference includes the operating-state dependence of the carrier response and the recurrent transfer. We compute its second-order source from the PING quadratic terms at the central carrier, 300 Hz, and apply the full linearized network response over 15–90 Hz in 0.025 Hz steps. Using the distinct AM sidebands changes the peak gain by less than 0.1% across this sweep.

The simulated coupling sweep uses the AM surrogate, $A_f = 8$, with 5 s settling and a 4 s window. We search around each linear resonance, extending the frequency grid until each sampled maximum is bracketed, and compare amplitudes $A_f = 2, 4, 8$ at nine couplings. Local parabolic interpolation changes the sampled peaks by less than 1.3%. The figure scales the linear reference to the median measured gain over $14.5 < J < 16$; the amplitude check instead normalizes peak response divided

by A_f^2 and linear gain to its value at $A_f = 2$, $J = 15.53$. These normalizations compare gain variation with the detector coefficient allowed to vary along this unpinned sweep.

We check the stability of the driven J-curve endpoint by integrating the full variational matrix along its trajectory. At carrier 300 Hz and beat 50.5 Hz, the AM input repeats every 2 s. After 8 s of settling, we compute the perturbation multipliers over that period and repeat over the next period. Moduli below one indicate decay of perturbations to the periodic response. Two-tone controls at $J = 16.5663$ and 16.5901, both with a 50.5 Hz beat, use 5 s settling, direct perturbation decay, and an 8 s finite-time Lyapunov estimate.

3.6 QIF spiking network (microscale of NMM2)

NMM2 is the exact infinite-population mean field of all-to-all QIF neurons [31, 26, 33]. We integrate that network directly to compare beat-locked and ongoing activity at the spike level: $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (22)), with single-neuron voltages V_j, finite spike threshold $V_{\text{peak}} = 100$ and reset $-V_{\text{peak}}$ in model voltage units, excitabilities $\eta_j \sim \text{Lorentzian}(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout.

The common background drive $\bar{\eta}$ places the field-free mean field below or above a gamma Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz). We use $\bar{\eta} = 0.85$ with a 55 Hz beat and $\bar{\eta} = 1.5$ with a 42 Hz beat. Both runs have carrier $f_c = 300$ Hz and amplitude $A_f = 35$ per tone. We label the conditions by their field-free state and separate the beat-locked component from other gamma activity. Spectra and rate folds resolve frequency components but do not establish phase capture.

We also run both conditions at $A_f = 10$ per tone with the same measurement settings. For $\bar{\eta} = 0.85$, separate checks double the population size, double the measurement window, or halve the integration step; further runs raise V_{peak} to 200 and 400 at both time steps. An equal-RMS mean-field control replaces the two tones by one 300 Hz carrier with peak amplitude $\sqrt{2}A_f$, using the same duration: both inputs have mean-square amplitude A_f^2 over these windows. The carrier comparison starts from the field-free equilibrium and uses 800 ms settling and 800 ms measurement, with a 3200 ms window to check spectral leakage at the larger amplitude.

For two nearby carriers, the fast-carrier approximation gives a constant shift of the excitatory drive, $\delta\eta_E \simeq A_f^2/(\omega_c\tau_E)^2$, from the quadratic voltage term. We refine this estimate using the full PING linear carrier responses and their second-order DC source, then solve for the carrier-averaged equilibrium and its stability, including the source's dependence on that equilibrium. This approximation isolates the mean shift; direct two-tone integrations retain the beat forcing and provide a separate check.

The network is integrated with Euler steps of 0.0025 ms, settled for 800 ms (over four field-free relaxation times at $\bar{\eta} = 0.85$, where the mean-field relaxation time is 176 ms) and measured over an integer number of beat cycles, 44 at 55 Hz and 34 at 42 Hz (800 and 810 ms), removing the phase-coverage bias from incomplete beat cycles. Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega/\langle r_E\rangle$ (a stable measure; the off→on lock-in ratio would divide by the field-free floor), and per neuron by the pairwise phase consistency of each neuron's spike phases at Δf (PPC; [45]), which is unbiased in the spike count, over neurons with at least ten spikes in the window.

The weak-drive beat depth depends on the finite spike cutoff: it ranges from 0.093 at $V_{\text{peak}} = 100$ to 0.121–0.141 at the two larger cutoffs across tested steps, against the mean-field value 0.128. The mean-rate increase stays at 2.8–2.9%. Population-size and time-step checks do not establish joint

convergence. At the stronger drive, rate modulation is less sensitive, but spike-phase concentration still depends on the cutoff.

3.7 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The JR sweeps vary coupling while keeping detection fixed. For each connectivity C, we solve for the external drive $p(C)$ that holds the field-free voltage at $v^* = 8.30$ mV, then obtain damping and modal frequency from the Jacobian. This fixes the background curvature without canceling stimulation-induced mean shifts.

The gain sweep uses the AM surrogate; the timing comparison uses genuine two tones and measures modulation and mean of the same pyramidal rate $r_P = \sigma(v + s)$. The tACS control uses a direct sinusoid at the same beat frequency and reads the same voltage lock-in as TI. Further controls match tACS to TI's generated beat source and repeat the comparison with longer settling and measurement windows. Waveforms, amplitudes, and run windows are recorded together in the supplement.

3.8 Vocabulary: forced response, entrainment, and gating

Forced response denotes the component driven at the beat Δf, quantified by the lock-in amplitude (Eq. (21)). A purely forced regime requires a stable operating state under stimulation; being below Hopf with the field off is insufficient. *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator [46]; in the maps we screen for it with the dominance criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency. This identifies frequency dominance, not phase capture. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [47, 48]. Capture and gating are dynamically different, and the same word would otherwise cover both. The QIF rate fold measures beat-locked firing, which does not by itself establish gating of gamma amplitude.

The coupling sweeps approach the Hopf from field-free stable foci ($\gamma > 0$), so the $1/\gamma$ law concerns forced susceptibility. The frequency-dominance maps on the oscillatory side describe a different observable: their bands broaden with drive, but an Arnold tongue describes phase capture of the autonomous limit cycle and requires phase tracking.

4 Results

Both the JR column and the exact QIF mean field generate a beat and amplify it near a Hopf bifurcation. The detector law fixes the local source and its spatial weighting; recurrent coupling sets the gain. The spiking network then distinguishes modulation of ongoing firing from field-induced gamma.

4.1 The JR column generates and resonantly amplifies the beat

An input containing only the two tones $f_c \mp \Delta f/2$ produces an output oscillation at their 11.1 Hz beat, near the column's alpha resonance (Fig. 2). The alpha peak is absent from the input spectrum (Fig. 4).

In the stable-focus regime ($p > p_H$), the response is approximately Lorentzian and its peak grows from 0.23 to 0.43 to 0.70 mV toward the Hopf (Fig. 5), as expected from the $1/\gamma$ susceptibility. Without a drive the focus is silent at Ω; the nonlinearity generates that component and recurrence amplifies it. The peak shifts with the driven operating state; the near-Hopf law gives its leading gain, not its exact frequency. The coupling sweep tests that law over a decade of damping. On the alpha branch below p_H, the column free-runs at f_0, so the lock-in can contain both driven and autonomous activity. The dominant output frequency f_{out} then identifies a frequency-dominance region (Fig. S4), without establishing phase capture.

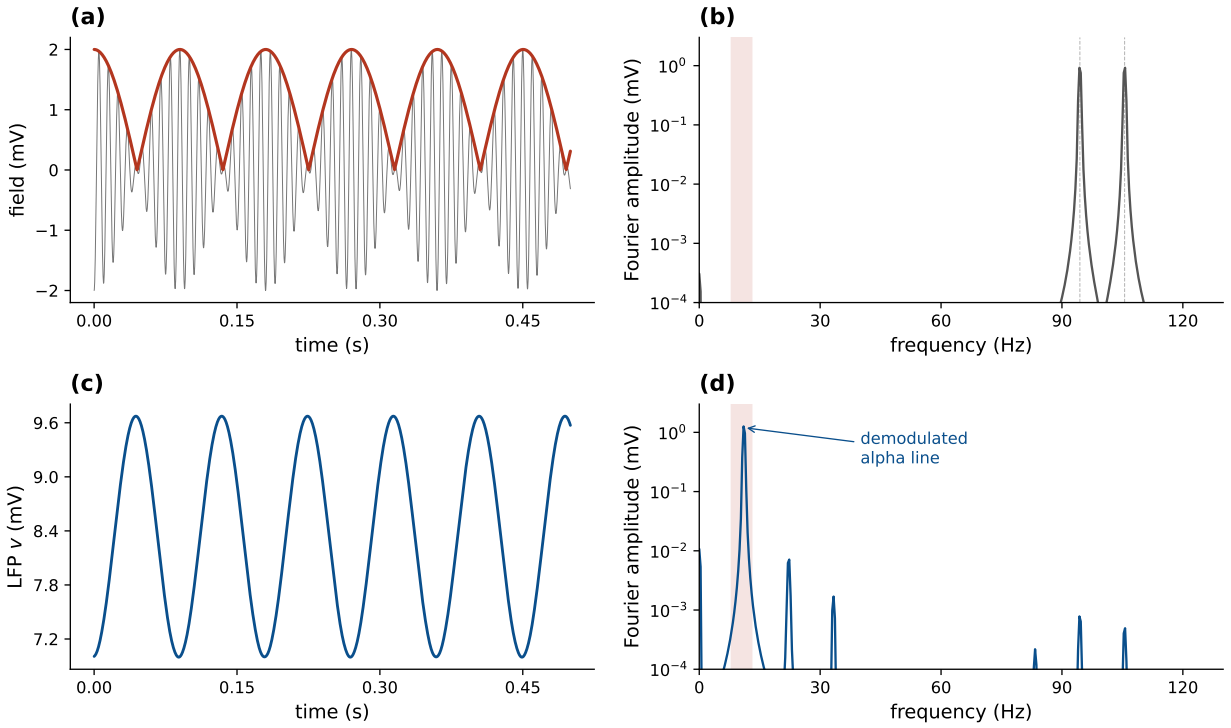


Figure 4: The JR column generates a beat absent from the input. *(a)* Two equal tones at 94.45 and 105.55 Hz and their envelope, $2\varepsilon|\cos(\Omega t/2)|$. *(b)* Input spectrum: only the two tones are present. The alpha-band floor is 1.6×10^{-8} of their amplitude, from finite-window leakage. *(c,d)* The synaptic voltage $v = y_1 - y_2$ and its spectrum show the generated 11.1 Hz beat. The field-free column is a stable focus near its alpha Hopf.

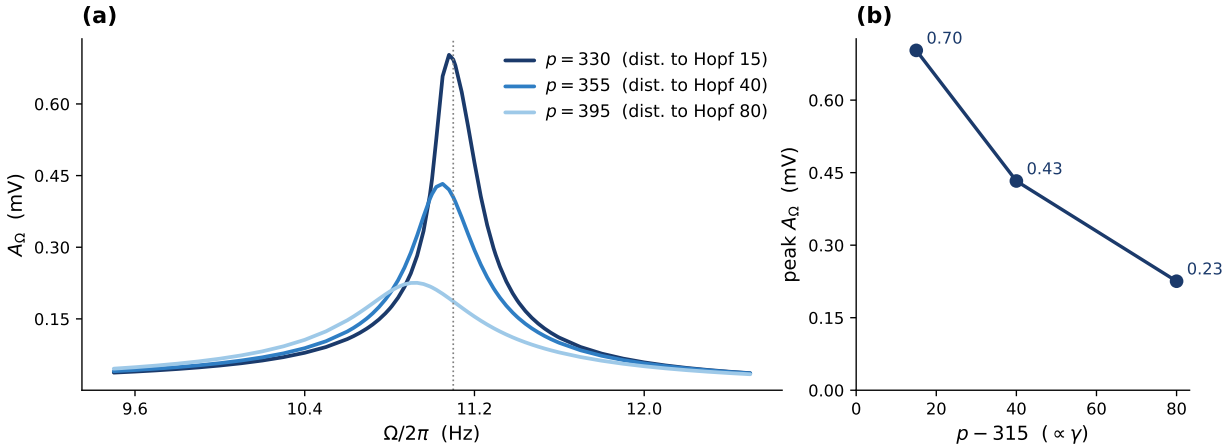


Figure 5: The JR beat resonance sharpens toward the Hopf. Stable-focus response at $\varepsilon = 0.3$ mV. *(a)* Lock-in amplitude against beat frequency; the finite-amplitude peak moves from 11.08 to 10.93 Hz across the curves. *(b)* Peak amplitude increases as p approaches the upper Hopf from above. These three points show sharpening and growth; the coupling sweep quantifies the $1/\gamma$ scaling and its saturation.

4.2 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 6). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner in the point-membrane model (Fig. S5). Its 16 ms response gives beat attenuation of about 10^{-4}–10^{-6} at 1–10 kHz; the cable-end case in the passive-cable calculation gives about 10^{-2}–10^{-3}. The assumed 0.2 ms point-membrane response retains appreciable carrier access.

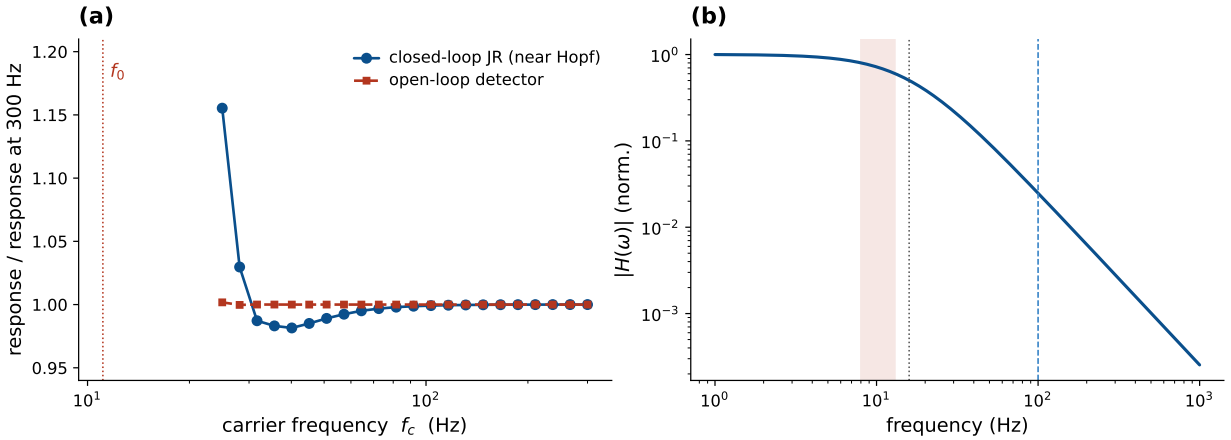


Figure 6: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is nearly flat across carrier frequency $f_c = 25$–300 Hz for the open-loop detector and for the closed-loop JR near the Hopf, with deviations as f_c approaches the natural band. Each curve is normalized by its own response at 300 Hz: 0.00670 s^{-1} open loop and 0.512 mV closed loop. Their ratio is a rate-to-voltage transfer of 76.4 mV s. The open-loop carrier independence follows to leading order from Eq. (6). *(b)* The second-order-synapse low-pass $|H_s(\omega)|/|H_s(0)|$ passes the envelope ($\sim$0.7) and suppresses the 100 Hz carrier ($\sim$2.5%). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

We also drive the full recurrent column with a kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant τ) for each modeled compartment before its polarization reaches the sigmoid. The column demodulates and resonates at the recovered alpha in the fast-access case ($\tau = 0.2$ ms), as Eq. (15) predicts (Fig. S9a). With two genuine tones centered at 2 kHz and beating at 10.914 Hz, the output contains the beat absent from the input (Fig. S6); the analytic access factor is $|H_m|^2 = 0.137$ for $\tau = 0.2$ ms. This quantifies the fast-access hypothesis under the specified response time and coupling.

4.3 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 7)—demodulation is the sigmoid-curvature term of Eq. (6), and the closed-loop operating points sit where σ'' is large and negative. Changing its sign reverses the source phase; at matched network susceptibility, its magnitude sets the same resonant-gain calculation on either flank.

In the recurrent column, the response scales quadratically at small drive (log–log slope 1.99, measured as in §3.2), and linearizing the field-receiving sigmoid reduces it from 1.34 to 0.0021 mV (Fig. 7b,c). The open-loop surrogate comparison in panel (d) supports its use for leading-order beat sweeps, with the waveform and near-Hopf limitations specified in §3.1.

4.4 Realistic-head comparison of envelope and quadratic maps

Changing the current ratio moves the envelope and mean-square foci in opposite directions in this head model, while the quadratic beat map only rescales. Figure 8 compares the three maps of Table 1, the envelope excursion $\mathcal{M}[\mathcal{A}]$, the quadratic excursion $\mathcal{M}[\frac{1}{2}\mathcal{A}^2]$ and the mean-square polarization $\mathcal{I}$, for the Violante montage in IXI002. All three use the same local polarizations, along the direction chosen at 1:1 and held fixed (§3.3).

At 1:1 ($2 + 2$ mA) the hippocampal means are 0.16 mV, 0.018 mV^2 and 0.0094 mV^2, respectively. The target-to-reference ratio is 1.22 for the envelope map, 1.17 for the quadratic excursion (1.08 degree-matched) and 0.92 for $\mathcal{I}$ (0.96). Once degree is matched, the quadratic map therefore favors

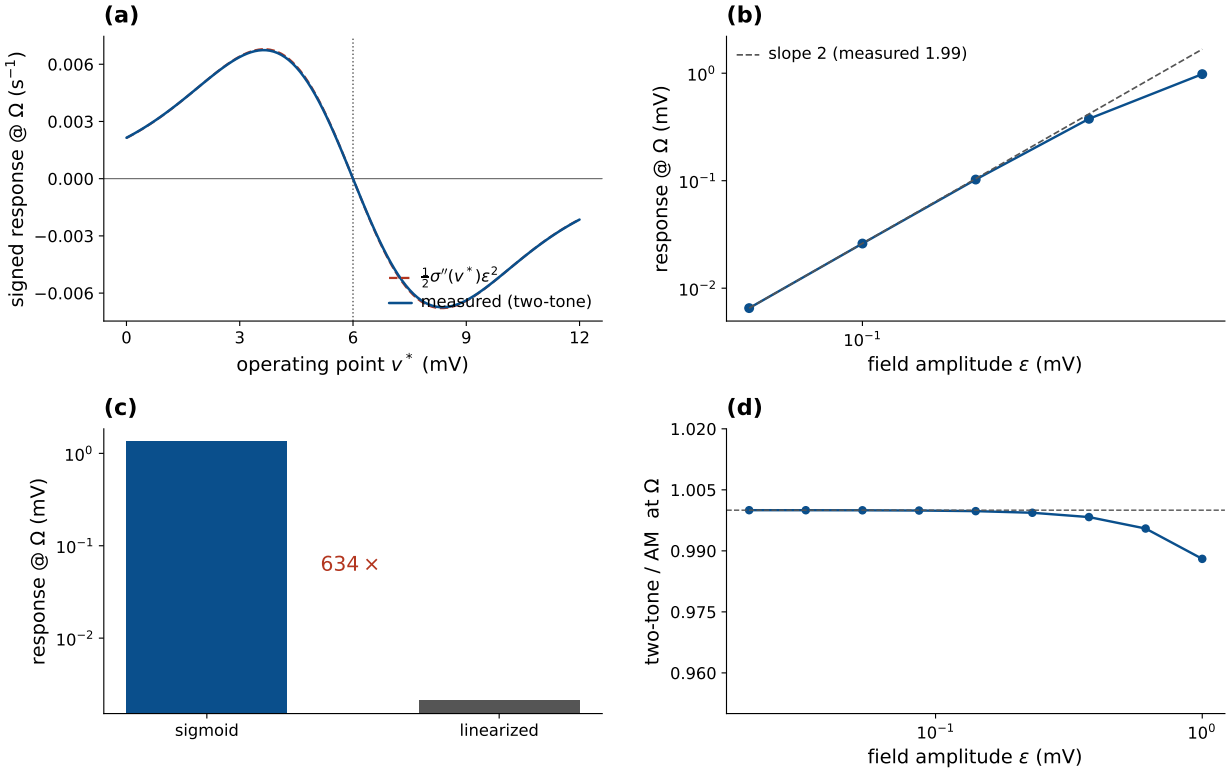


Figure 7: Curvature determines the JR detector. *(a)* The signed open-loop beat follows $\sigma''(v^*)\varepsilon^2/2$ within 0.9%, vanishing and reversing sign at v_0. *(b)* The recurrent response has amplitude slope 1.99. *(c)* Linearizing the receiving sigmoid reduces the beat 634-fold. *(d)* Open-loop two-tone/AM beat ratio at $v^* = 8$ mV: 1.0000 at 0.02 mV and 0.988 at 1 mV. This checks the leading local source, not the nonlinear recurrent response near the Hopf.

the hippocampus less than the envelope map does, as Eq. (9) predicts where the dominant local polarization is larger in the reference than at the target; $\mathcal{I}$ favors the reference.

Changing the current ratio separates the three maps (Fig. 8a–c,f). From 1:1 to 1:3 the quadratic excursion is multiplied by $I_1^{\text{pair}} I_2^{\text{pair}}/(2\text{ mA})^2 = 0.75$ at every node, to machine precision. Its normalized map, its target-to-reference ratio and its focus do not change: this is the fixed-state product law of §2.3. The envelope focus moves 2.5 mm anteriorly (2.2–4.0 mm over the 5–50% fractions), toward the pair whose current falls. Its ratio drops from 1.22 to 0.95, and its hippocampal mean falls by 45%. The focus of $\mathcal{I}$ moves 4.5 mm posteriorly (2.4–8.2 mm), toward the pair whose current rises. Its hippocampal mean rises by 24%, while its ratio changes little (0.92 to 0.90).

The sweep shows two kinds of steering (Fig. 8e,f). The envelope ratio peaks near balance (1.24 at a posterior share of 0.525) and falls to 0.89–0.94 at the extremes. Its focus moves only while the two carriers are comparable inside the hippocampus. At a posterior share of 0.6 or more, the anterior pair is the weaker carrier at more than 96% of hippocampal nodes; at 0.4 or less, the posterior pair is. The envelope map then equals twice the weaker pair's polarization over most of the ROI, and its measured focus changes little with further imbalance. The focus of $\mathcal{I}$ moves continuously instead, from 1.5 mm anterior at a posterior share of 0.1 to 6.3 mm posterior at 0.9, while its ratio stays between 0.79 and 0.94 (0.89–0.97 degree-matched). The quadratic excursion keeps one ratio and one focus across the whole sweep.

The isotropic beat map (Fig. 8d,g) has a target-to-reference ratio of 1.18 (1.08 degree-matched), close to that of the fixed-direction quadratic map. It also scales with $I_1^{\text{pair}} I_2^{\text{pair}}$, so the current ratio cannot steer it either.

The atlas label lowers the absolute ratios (envelope 1.17, quadratic excursion 1.04 at 1:1) but keeps every direction of focus displacement: envelope 2.0–3.8 mm anterior, $\mathcal{I}$ 3.2–5.9 mm posterior, quadratic excursion zero. Along the radial direction the displacements also keep their directions (envelope 1.9–2.8 mm anterior, $\mathcal{I}$ 2.0–3.7 mm posterior), but the ratios change more than with the label: 1.62 for the envelope and 1.97 for the quadratic excursion at 1:1. In this subject the assumed orientation, more than the label, sets the absolute contrast. Across all 32 combinations of montage, label, ROI tissue definition and focus fraction, every envelope shift is anterior (0.7–8.1 mm), every shift of $\mathcal{I}$ is posterior (0.5–10.6 mm), and every shift of the quadratic excursion is zero.

All four maps are geometric quantities before detector gain. $\mathcal{I}$ is nonnegative and does not by itself assign excitation or suppression (§2.5). These results come from one subject and one montage, with quasi-static conductivities and a uniform coupling.

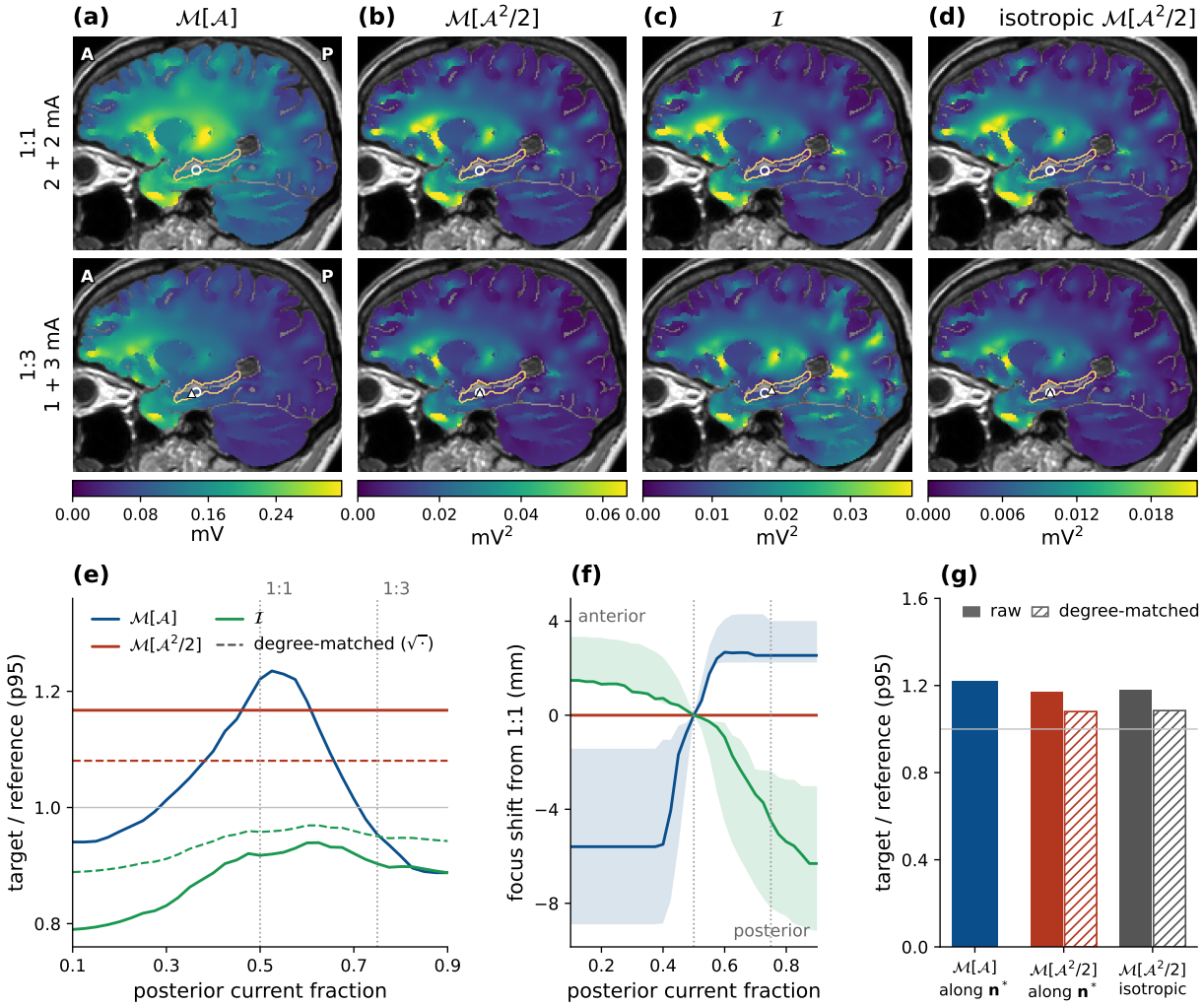


Figure 8: Envelope and quadratic maps in IXI002. Violante montage; left hippocampus outlined in gold, anterior to the left. *(a–d)* Sagittal maps at 2 + 2 mA (top) and 1 + 3 mA (bottom): envelope excursion, quadratic excursion, mean-square polarization, and isotropic beat. The first three use the fixed 1:1 envelope-maximizing direction. Each color scale ends at its 1:1 gray-matter 99th percentile. Circles and triangles mark the projected top-10% focus at 1:1 and 1:3. *(e)* Target-to-reference contrast; dashed curves use square-root maps to match degree. *(f)* Anterior focus displacement: top 10% (line), range over 5–50% (band). *(g)* Contrasts at 1:1, raw (solid) and degree-matched (hatched). The reference is gray matter outside the ROI. All maps precede detector gain: $\mathcal{I}$ weights $Y_{\mathrm{DC}} = \sigma''\mathcal{I}/2$, but does not set its sign.

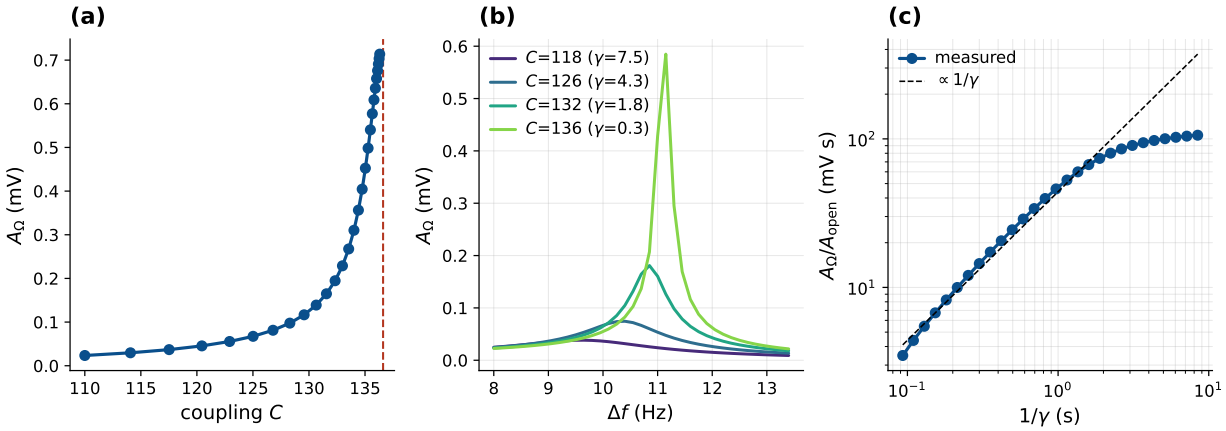


Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit). The input is the AM surrogate of Eq. (20). The field-free operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151\ \mathrm{s^{-1}mV^{-2}}$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling C approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain $A_\Omega/A_{\mathrm{open}}$, in mV s, follows $1/\gamma$ (dashed) and saturates near the bifurcation.

4.5 Coupling sets the gain at fixed detection

Changing connectivity at a fixed field-free operating point isolates recurrent gain from detection. We vary C and adjust the external drive $p(C)$ to hold v^*, its curvature $\sigma''(v^*)$, and the open-loop beat amplitude $A_{\mathrm{open}} \simeq \frac{1}{2}|\sigma''(v^*)|\varepsilon^2 m$ constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows $\sim$30$\times$ and the envelope resonance sharpens (Fig. 9a,b), a network effect since the local detector curvature is fixed while the recurrent transfer changes. Its modal frequency and input–output projections vary with C as well as its damping. The rate-to-voltage gain $A_\Omega/A_{\mathrm{open}}$ (mV s) tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input p rather than the coupling C gives a similar leading gain–γ trend. Both paths approach a weakly damped mode; their finite-distance gains need not coincide because the modal projections can differ.

4.6 The resonance amplifies the envelope-locked modulation more than the mean rate

Pinning v^* to hold the local detector curvature fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, we read both quantities from the same observable, the pyramidal rate $r_P = \sigma(v + s)$ with the field in its argument. With genuine two tones, its beat-locked modulation at $\Delta f = f_0$ grows 13.2$\times$ (26 to 339 mHz), tracking $1/\gamma$ before saturating; its mean shift grows 4.4$\times$ (-1.8 to -7.7 mHz), but stays below 0.2% of the 3.9 Hz baseline (Fig. S11). The ratio of modulation to mean-shift magnitude rises from 14.5 far from the Hopf to 58 near $\gamma = 1\ \mathrm{s^{-1}}$, then falls to 44 closest to the Hopf as rectification of the amplified oscillation accelerates the mean shift. The surrogate, whose direct DC source is 3/4 of the two-tone value, gives ratios of 19, 66, and 45. A small mean-rate change can therefore accompany a much larger beat response in this weakly driven regime, consistent with primate observations [20]; both effects are compared in the same rate units.

The demodulator feeds the network a constant term and a line at the beat Ω (Eq. (5)), and the transfer $G(\Omega)$ of Eq. (13) treats them differently: $|G(\omega_0)| \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $|G(0)| \propto 1/(\omega_0^2 + \gamma^2)$ stays finite. This comparison describes the response to the detector's DC and beat sources. The growing beat response itself rectifies and contributes another mean shift, quadratic in its amplitude. Before saturation that contribution can grow as $1/\gamma^2$, so bounded $G(0)$ does not imply a bounded mean shift throughout the approach to the Hopf.

4.7 TI and tACS share the linear amplifier

TI and direct low-frequency stimulation share the same linear amplifier. Matching their generated in-band drives therefore matches their linear outputs; resonance alone gives TI no efficiency advantage. For the AM surrogate with amplitude $\varepsilon_{\mathrm{TI}}$ and modulation depth m, and a tACS input of amplitude $\varepsilon_{\mathrm{tACS}}$, the ratio of voltage lock-in amplitudes in linear response is

$$\mathcal{R}_{\mathrm{lin}} = \frac{|\sigma''(v^*)|\,\varepsilon_{\mathrm{TI}}^2 m}{2\sigma'(v^*)\,\varepsilon_{\mathrm{tACS}}}.$$

The common network transfer cancels, so the ratio has no dependence on γ when both responses are linear about the same state. At the amplitudes used in Figs. 9 and S3, $\mathcal{R}_{\mathrm{lin}} = 0.143$. The surrogate sweeps give approximately 0.14 far from the Hopf, rising to 0.24 at $\gamma = 1.07$ s^{-1} and 0.47 at 0.12 s^{-1}. Genuine two tones give 0.14, 0.24, and 0.46, respectively, a change of at most 2.2% across the grid. These are response ratios for the stated unequal input amplitudes, not an equal-power comparison. In controls at three couplings, matching the tACS input to TI's leading beat source gives TI/tACS output ratios of 0.99, 0.94, and 0.92 for the surrogate, and 0.99, 0.93, and 0.90 for two tones. The stronger direct response saturates earlier near the Hopf: multiplying its input by 6.99 increases the output by only 6.97, 3.95, and 1.96 at these three couplings. Stimulation-induced mean shifts, which field-free pinning does not cancel, account for this departure from the linear ratio.

4.8 Exact mean-field confirmation: the NMM2 PING gamma generator

The exact PING mean field generates and amplifies the beat through its derived v_E^2 nonlinearity. With a 300 Hz two-tone field entering the excitatory current, the lock-in of r_E shows a growing gamma resonance below the Hopf and a response superposed on autonomous gamma above it (Figs. 11, S8). The resonance ridge intensifies through the Hopf; its appreciable frequency shift with operating point makes it diagonal, unlike the JR sweep shown here [33].

This derived detector lies on the fast side of its crossover at 300 Hz: carrier voltage excursions mix through v_E^2 before the rate reads the result. For the full eight-state PING (Eq. (22)), each high-frequency current tone of amplitude I_j produces a voltage phasor $I_j/(i\omega_j\tau_E)$. The v_E^2 term therefore supplies a difference-frequency source of magnitude $I_1 I_2/(\tau_E^2\omega_1\omega_2)$ inside the equation for $\tau_E\dot{v}_E$. After division by τ_E, this source is propagated through the network susceptibility at Ω. At four stable operating points, the resulting response agrees with the full second-order calculation to within 1.1% at 300 Hz, 0.1% at 1 kHz, and 0.03% at 2 kHz (Fig. 10b).

In NMM2 PING, J is the mean inter-neuron QIF synaptic weight. Sweeping J at fixed background drive toward the gamma Hopf reveals two competing modes (Fig. S10). A second mode near 55 Hz carries the linear-response peak over $14.6 \lesssim J \lesssim 16.2$; the critical mode carries it at lower couplings and again near the Hopf, where its frequency is about 50 Hz. In that near-Hopf interval, $0.0014 \lesssim \gamma \lesssim 0.0081$ ms^{-1}, the linear gain scales as $1/\gamma$ with a fitted exponent of 1.02 over a 5.8-fold damping range. The full-sweep exponent is 0.71, so a single-mode law does not describe the whole curve. This refines the coupling-controlled sensitivity expected in QIF networks [33]: both coupling and rectifier are derived, but the dominant mode can change.

At $A_f = 8$, the simulated response exceeds the normalized linear reference by up to 48% in the amplitude check, then saturates. At a quarter of that amplitude ($A_f = 2$, one sixteenth of the power), the departures stay within 11% at the nine tested couplings. The fully forced endpoint remains attracting under both waveforms, even where a DC-only approximation predicts instability; γ describes the field-free mode. The gain reference includes the state-dependent detector. The alpha JR and gamma NMM2 circuits share the $1/\gamma$ gain scaling of their critical mode despite their different mixers, with amplification adjustable through coupling.

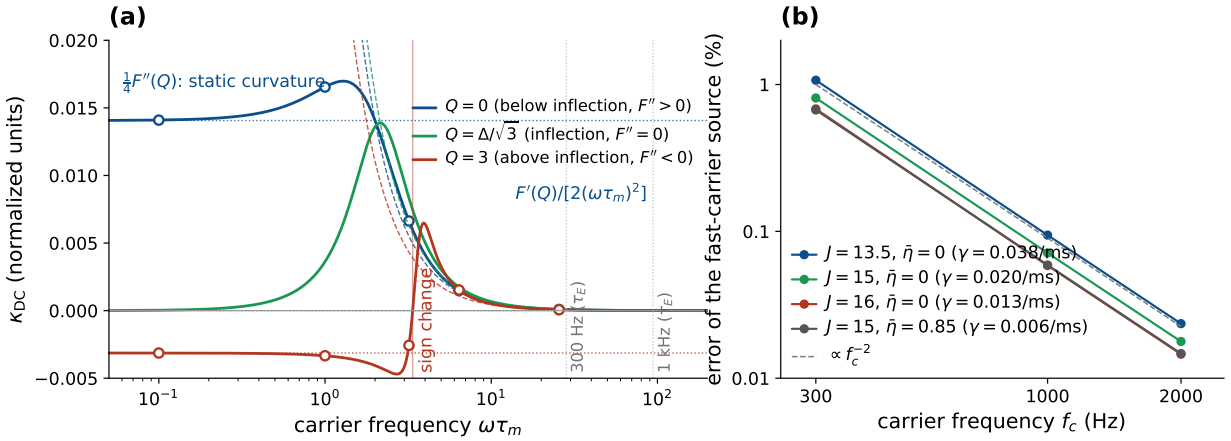


Figure 10: Static curvature at slow carriers, slope at fast ones. *(a)* Second-order DC rate shift of an uncoupled MPR population ($\Delta = 1$) under a current carrier, Eq. (6), for operating points below, at and above the inflection of $F(Q)$. Dotted: the static limit $\frac{1}{4}F''$; dashed: the fast-carrier limit $F'/[2(\omega\tau_m)^2]$; circles: RK4 integrations at $I_{ac} = 0.05$ averaged over whole carrier periods (relative error below 10^{-3}). Above the inflection the sign changes with carrier frequency. The horizontal coordinate is $\omega\tau_m$; the carrier markers use $\tau_m = \tau_E = 15$ ms. *(b)* PING of Eq. (22): relative error of the fast-carrier difference-frequency source against the exact second-order response at four stable operating points (γ the damping of the gamma mode), two tones centered at 300, 1000 and 2000 Hz.

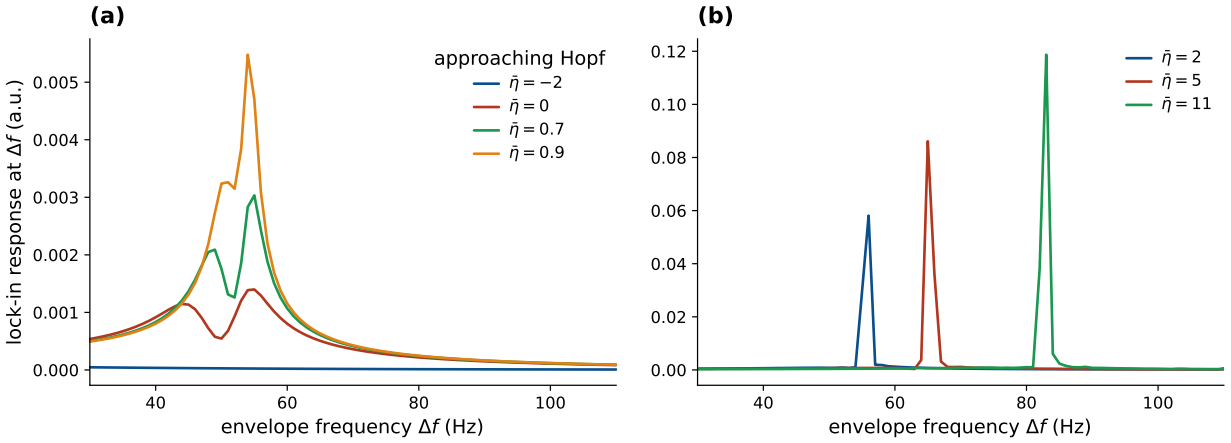


Figure 11: NMM2 PING gamma resonance (exact mean field, genuine two tones). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_H = 1.110$) forced resonance at the gamma frequency ($f_0 \approx 54$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ ms^{-1} as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_H$) response in the autonomous-gamma regime, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity, with no sigmoid anywhere in the model. Matched AM and two-tone drives agree at Ω to 0.07% well below the Hopf and to 4% at $\bar{\eta} = 0.9$, consistent with the JR surrogate control (Fig. 7d).

4.9 QIF responses distinguish beat modulation from field-induced gamma

Below the field-induced Hopf, the QIF network shows beat-locked modulation with a small mean-rate change; above it, the field induces gamma and adds a beat component. The 8000-neuron populations test this distinction at the spike level [31, 26, 33] (Figs. 12, 13).

The equal-RMS unmodulated carrier provides a control for this distinction in the mean field (Fig. 13c,d). It produces a rectified E-drive shift within 3% of TI's and a similar mean-rate increase. At $A_f = 35$, both waveforms induce gamma near 51 Hz, whereas TI also produces the 55 Hz beat component. At $A_f = 10$, neither induces gamma, and TI alone produces a resolved

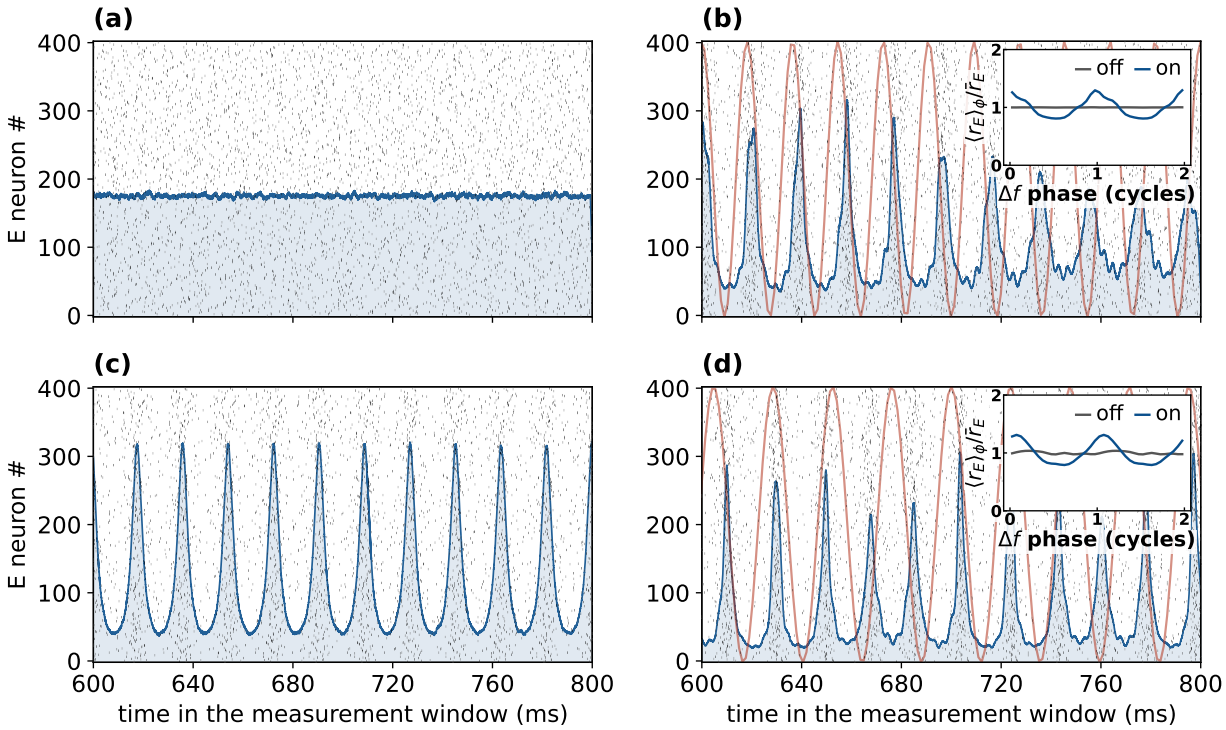


Figure 12: Beat-locked activity superposed on gamma. QIF network, $N = 8000$ per population, $A_f = 35$ per tone, carrier 300 Hz. Black ticks: 400 E neurons; blue: smoothed rate on its own scale; red: envelope in arbitrary units. The last 200 ms are shown. *(a,b)* Field off/on at $\bar{\eta} = 0.85$, beat 55 Hz: induced gamma near 52 Hz dominates. *(c,d)* Field off/on at $\bar{\eta} = 1.5$, beat 42 Hz: autonomous gamma remains dominant. Insets fold rate on beat phase over 44 and 34 cycles, respectively, normalized to each condition's mean (off gray, on blue; scale 0–2). Folding suppresses non-beat-locked gamma and exposes the smaller beat component; it does not establish gamma-amplitude gating or phase capture.

beat response. A longer window reduces the single carrier's apparent 55 Hz depth from 0.011 to 9.2×10^{-5}, while TI's stays at 0.30–0.31; the former is spectral leakage. The equal-power control separates induction of gamma from the beat response and has been tested only in the mean field.

The $A_f = 35$ runs are finite-amplitude illustrations: the fast-carrier estimate gives a voltage excursion $A_f/(\tau_E \omega_c) \approx 1.2$ per tone in the model's voltage units. Its rectified excitatory-drive shift is about 1.5, compared with 0.12 at $A_f = 10$. At $\bar{\eta} = 0.85$, the Hopf threshold for a drive shift applied only to the excitatory population is $\delta\eta_E^* \approx 0.286$. The larger field is therefore expected to cross this threshold, whereas the smaller one remains below it.

At $\bar{\eta} = 0.85$ and tone amplitude $A_f = 10$, the spiking network has a beat depth of 0.093 at 55 Hz and a mean-rate increase of 2.8%: modulation exceeds the fractional mean-rate change by a factor of 3.4. The mean field gives a larger depth, 0.128. The response depends on the finite spike cutoff, and quantitative convergence remains unestablished at this population size (§3.6). At $\bar{\eta} = 1.5$, the corresponding 42 Hz depth is smaller, 0.023, against a 1.6% mean-rate increase, and the existing gamma dominates the raster.

At the larger amplitude $A_f = 35$ and $\bar{\eta} = 0.85$, stimulation produces a dominant line near 52 Hz, with modulation depth about 0.70, and a smaller component at the 55 Hz beat, with depth 0.29. The second-order carrier-averaged system predicts an excitatory drive shift $\delta\eta_E \approx 1.65$, beyond the E-only Hopf threshold $\delta\eta_E^* \approx 0.286$, with a leading eigenvalue whose real part is $+0.032$ ms^{-1}. Direct mean-field integration reproduces the off-beat line, supporting the interpretation of panel (b) as field-induced gamma with a smaller forced component.

At $\bar{\eta} = 1.5$, gamma activity is already present without stimulation. With the field on, the rhythm

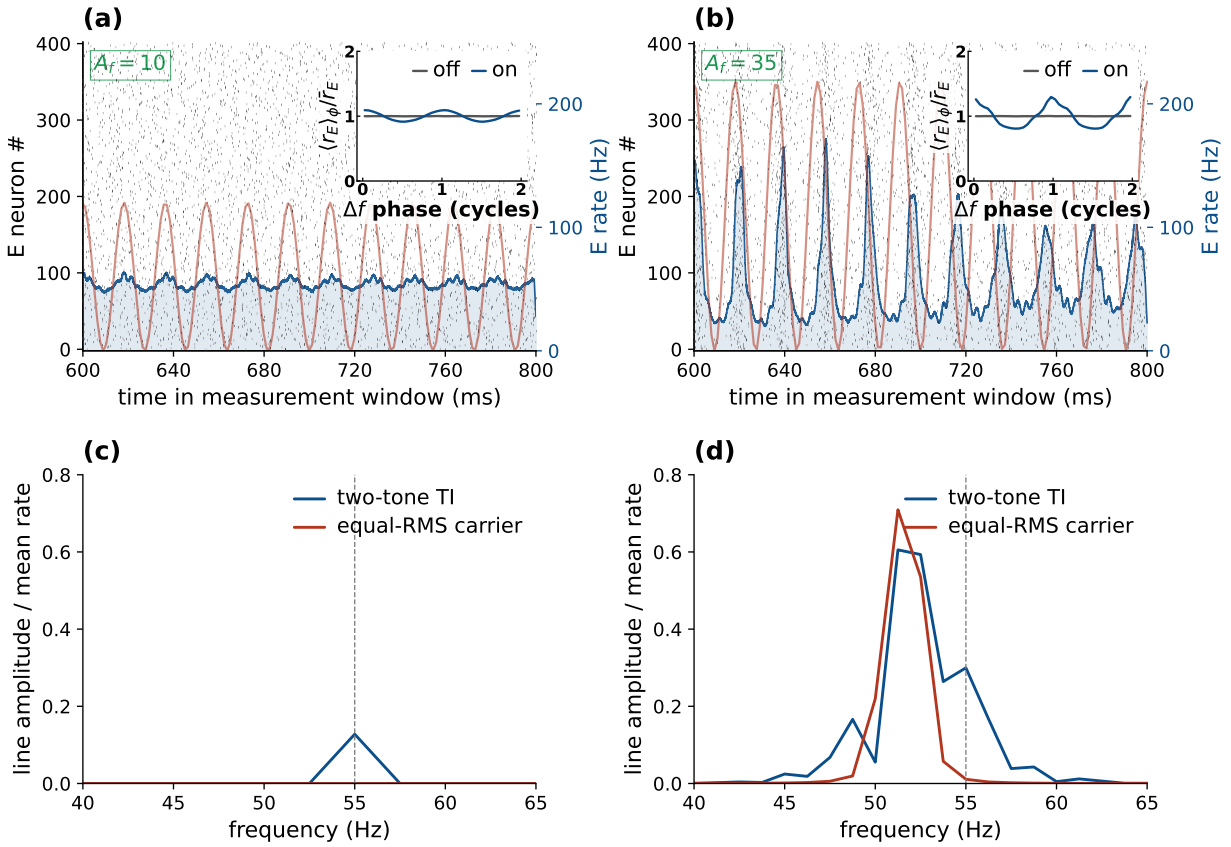


Figure 13: TI and an equal-power carrier across the gamma threshold. Background $\bar{\eta} = 0.85$, carrier 300 Hz, beat 55 Hz. *(a,b)* Spiking network at $A_f = 10$ and 35 per tone. Black: 400 of 8000 E neurons; blue: population rate on a common scale; red: envelope. Insets show mean-normalized phase folds (off gray, on blue). Beat depths are 0.093 and 0.290, with mean-rate increases of 2.8% and 10.5%; the larger input also induces gamma. *(c,d)* Mean-field spectra under TI (blue) and one carrier of peak amplitude $\sqrt{2}A_f$ (red), normalized to mean rate. Both induce gamma at the larger input; TI adds the 55 Hz beat (dashed). Hann windows span 800 ms after 800 ms settling. Spiking panels use $V_{\text{peak}} = 100$; weak-drive depth remains cutoff-sensitive.

near 53–54 Hz remains about four times larger than the 42 Hz beat-locked component. Gamma therefore dominates the raster in panel (d); the coexistence of the rhythms produces slower variations in burst height. Folding the rate on the beat phase suppresses the non-beat-locked gamma and exposes the smaller component. The inset shows that phase average, not the full rate variation.

Mixing of slow and fast inputs can generate endogenous amplitude modulation [25]; here the recovered beat mixes with the ongoing gamma. The spectrum has sidebands near 95 and 11 Hz (Fig. S12b), establishing nonlinear mixing without distinguishing gamma-amplitude gating from phase modulation; field-free gamma frequencies agree between the spiking network and mean field (panel a).

At $A_f = 35$, the mean rate rises by about 8–11% across the two conditions, while the beat modulation depth, $A_\Omega / \langle r_E \rangle$, rises from below 0.003 to 0.29 ($\bar{\eta} = 0.85$) and 0.23 ($\bar{\eta} = 1.5$) (Fig. S12c). Modulation is about three times the relative mean-rate change, but per-neuron locking is weak: the pairwise phase consistency of single-neuron spike phases at Δf rises from its field-free control (median -0.02) to a median of 0.003–0.005, and the pooled vector strength from below 0.02 to 0.12–0.13. The population's beat-locked component is built from many weakly biased cells; these finite-amplitude runs do not establish its size under weak physiological stimulation.

5 Discussion

5.1 Cellular access and mixing require separate calibration

Access sets the local polarization, detection generates the beat, and the circuit sets its amplification. Esmaeilpour et al. [6] already related network TI sensitivity to short, axon-like membrane times and distinguished sensitivity from selectivity mediated by network adaptation. Here the exact QIF mean field connects static and dynamic mixing, the weak-input product law determines spatial weighting, and matching the generated drive exposes the linear amplifier shared by TI and tACS.

At fields of order 1 V/m, channels respond through membrane polarization, well below the fields associated with direct protein effects [49].[4] Presynaptic terminals can polarize more strongly than somata [50, 51, 52], but small size alone does not imply fast charging: specific membrane resistance times capacitance is area-independent, while conductance, geometry, and stimulation mode determine the response.

Short access times motivate the model but remain preparation-dependent. The 1 and 0.4 ms cases of [6] precede our effective-time hypothesis. Luff et al.'s pulse-width-modulated TI model uses 1 ms within a 1–100 ms passive-time sweep [53]. The 0.7 μs charging measured by Semenov et al. under strong microsecond field pulses [23] concerns a different response mode from intracellular current charging or periaxonal conduction. A 10–100-fold reduction in QIF membrane time is therefore a sensitivity hypothesis, whose population consequence follows from the slow-synapse limit without adding a periaxonal circuit.

The quadratic law specifies the leading mixing operation, not its coefficient. QIF derives one realization; cellular mixing measured or modeled in single neurons [2, 10] can feed the recurrent amplification seen in slices and simulations [16, 17]. These are complementary stages. A static curve describes mixing only when population relaxation is fast relative to the carrier as well as the synapses. Otherwise the coefficient retains its own frequency dependence, while slope converts slow input into rate and enters recurrent gain.

5.2 Detector laws predict different spatial responses

A spatial TI dose map depends on this physiology. The usual envelope-modulation map assumes a response determined by envelope excursion; the square-law map instead weights it by the larger local polarization (Eq. (9)). The accompanying steady source follows the summed carrier intensities, weighted by the local detector coefficient. Under uniform conductivity, access, and detection, its largest magnitude and the ideal envelope mean lie on compartment boundaries. The beat can instead peak inside. Orientation, access, operating point, and network susceptibility further shape the response. A sharp rectifier can approximate envelope following over a finite amplitude range; smoothing or heterogeneity can restore quadratic behavior at smaller inputs. The amplitude and harmonic tests in §3.2 distinguish these regimes.

Violante et al. [37] demonstrated envelope steering and a change in hippocampal physiology using different measurements. Their field models and cadaver recordings supported anterior envelope steering when the current ratio changed from 2 + 2 mA (1:1) to 1 + 3 mA (1:3) through the same electrode pairs. In participants, stimulation was confined to the 32 s encoding stage of each block. The 1:3 condition reduced the encoding-evoked blood-oxygen-level-dependent (BOLD) response relative to sham ($P = 0.006$) and 1:1 ($P = 0.015$); 1:1 did not differ significantly from sham ($P = 0.953$). The reduction was nominally largest anteriorly, but the stimulation-by-region interaction was not significant ($P = 0.76$). The voxelwise sham–1:3 comparison found significant reductions in 30% of anterior, 25% of posterior, and 2% of middle hippocampal voxels. Significant BOLD reductions thus extended beyond the anterior envelope focus.

[4] The review describes direct effects at 10 kV/m to GV/m, depending on molecule and protocol. It excludes fields below 0.1 kV/m, where it expects direct effects to be negligible against thermal noise.

All three geometric maps change between these current conditions, so a condition difference alone cannot identify the detector. In our IXI002 calculation (§4.4, Fig. 8), the envelope focus moves 2.5 mm anteriorly, whereas the quadratic beat retains its pattern and falls to 3/4 of its value everywhere at fixed neural state. The mean-square polarization changes by $\mathcal{I}_{1:3}/\mathcal{I}_{1:1} = (\mathcal{P}_1^2 + 9\mathcal{P}_2^2)/[4(\mathcal{P}_1^2 + \mathcal{P}_2^2)]$, exceeding one only where $|\mathcal{P}_2| > \sqrt{3/5}|\mathcal{P}_1| \approx 0.775|\mathcal{P}_1|$. Its hippocampal mean rises by 24% and its focus moves 4.5 mm posteriorly. Equal summed currents therefore do not match local RMS polarization.

At spatially uniform, fixed curvature, the 24% increase in hippocampal mean-square polarization predicts a 24% increase in the magnitude of the steady rate source $Y_{\mathrm{DC}} = \sigma''(v^*)\mathcal{I}/2$. Curvature sets its sign; recurrence determines the final rate response. By comparison, the hippocampal mean envelope excursion falls by 45%, and both the projected and isotropic quadratic beat maps fall by 25%. A response linear in the quadratic beat thus decreases; its own rectified mean shift scales as $(3/4)^2 \approx 0.56$ at fixed state in the weak-response limit (§4.6). Of these hippocampal map averages, only the mean-square polarization increases under 1:3, consistent with a larger magnitude of BOLD reduction if the steady source lowers activity and the hemodynamic response preserves that ordering. A slow, approximately linear hemodynamic readout suppresses the zero-mean 5 Hz beat at first order while retaining a slow shift over the stimulation window. Such a shift can add to the encoding-evoked response or alter its operating point; nonlinear neural or vascular responses can also convert the beat into a mean change.

Locally, the envelope excursion rises where $|\mathcal{P}_1| > 2|\mathcal{P}_2|$, reaching 1.5× where $|\mathcal{P}_1| \geq 3|\mathcal{P}_2|$, allowing a spatially selective or thresholded envelope response, and the nonsignificant 1:1–sham difference does not establish a null effect. Hippocampal curvature and recurrent and hemodynamic gains are uncalibrated; heterogeneous curvature can change the mean-source ratio. BOLD and functional connectivity measure neither source directly, so interpreting their spatial pattern requires a neural and hemodynamic readout model. A spatially resolved beat measurement at controlled state would test envelope steering against the fixed quadratic pattern; an equal-power unmodulated-carrier control would isolate effects of the steady source under the second-order matching conditions of §2.5.

5.3 Field-induced oscillations and resonant amplification

At higher intensities, TI and unmodulated kHz fields can have similar excitation thresholds, with modulation patterning firing rather than lowering the threshold [21, 23]; some peripheral responses do not require envelope extraction [22]. The weak-input calculation instead describes modulation of ongoing activity [20, 54], excluding direct carrier-evoked spiking and conduction block. The finite-amplitude QIF examples show a related circuit transition: the mean shift crosses the Hopf, inducing gamma alongside the beat response. Its boundary depends on both field amplitude and the initial distance from the bifurcation. These examples and the laminar extension go beyond the weak-input expansion without establishing physiological thresholds.

The recurrent amplifier is shared with direct low-frequency stimulation. Our coupling sweep fixes the JR field-free operating point and its local curvature while changing recurrent gain. A direct in-band drive shows the same $1/\gamma$ growth and resonance sharpening (Fig. S3), consistent with the known resonant susceptibility of recurrent excitatory–inhibitory networks [15, 33]. Near the Hopf, this is temporal critical slowing in a mean-field model, not evidence of a diverging spatial correlation length. Related models already reproduce tACS entrainment and AM-driven phase locking [55, 56, 57]. When TI and tACS are matched for effective in-band input, observable, and operating state, they share the linear amplifier. Within linear response, resonance alone therefore gives TI no efficiency advantage. At finite amplitude the response ratio can change near the Hopf, as the coupling sweep demonstrates (§4.7); its value then depends on drive amplitude and the driven state as well as on field-free damping.

A reverberation time $\tau_{\mathrm{rev}} = 1/\gamma$ gives a near-Hopf quality factor $\mathcal{Q} = \pi f_0 \tau_{\mathrm{rev}}$ and power bandwidth

approximately $1/(\pi\tau_{\mathrm{rev}})$. Population relaxation over a few hundred milliseconds [58] would give a quality factor of order ten at 10 Hz and a bandwidth of order 1 Hz. This is an order-of-magnitude comparison: the measured relaxation is not specific to the alpha mode. Narrow amplification also makes detuning consequential.

5.4 Efficiency depends on the generated drive

Absolute response estimates depend on access and transduction. Measured DC field-to-voltage couplings are 0.12–0.18 mV per V/m in hippocampus (0.21 at 10 Hz AC) [41, 42] and reach about 0.5 mV per V/m in cortical L5/6 pyramidal cells [40]; human cortical fields near 0.8 V/m have been reported at 2 mA [59, 60]. These static couplings do not fix the coupling at kHz. For equal TI carrier polarizations ε, the JR beat drive is $\sigma''\varepsilon^2/2$, whereas a direct drive of the same peak polarization gives $2\sigma'\varepsilon$. Their leading amplitude ratio is $\mathcal{R}_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$: the relative efficiency decreases as the input is weakened in this perturbative regime. A larger recurrent gain amplifies both drives.

For an illustrative fast-access budget, take a per-tone field $E_j = 0.5$ V/m, $\lambda_0 = 0.5$ mV per V/m, $f_c = 2$ kHz, and the assumed $\tau_m = 0.2$ ms. Then $\varepsilon = \lambda_0 E_j |H_m| \approx 0.092$ mV, rounded to 0.1 mV for scale; this is an assumed coupling budget, not an inferred polarization in the recorded cells. At $v^* = 8.30$ mV the JR detector gives $\mathcal{R}_{\mathrm{det}} \approx 0.8\%$ for 0.1 mV, for the generated drive relative to a matched-peak direct input. The smooth JR detector generates a small beat under this budget. Recurrence amplifies both TI and matched direct drives, preserving their linear-response ratio. Testing the estimate requires a polarization-calibrated sweep with the same readout; a measured PLV ratio alone does not supply it.

For comparison, the carrier-averaged, matched-peak, open-loop drive ratio has a numerically estimated supremum of 0.348 over bounded nondecreasing memoryless transfers (Eq. (12)). The positive-mixture-of-steps argument extends the logistic result to this class. Approaching it requires an active transition sharp relative to the input; for the logistic family at the illustrative polarizations, this means microvolt widths. The limiting detector is step-like and has a second-harmonic ratio near 0.15, distinct from both weak square-law detection and linear envelope following. This bound does not cover the dynamic QIF detector or all cellular mechanisms. A larger common matched polarization does not exceed the class bound; unequal actual polarizations can instead invalidate the matching. The unequal-current TI ratio and the matched-depth AM-tACS control do not measure this matched-peak, open-loop ratio. Thus approaching the ceiling would require leaving the smooth weak-input regime used here; within that regime, local curvature determines the generated drive.

The weaker TI effects reported by Vieira et al. [20] are compatible with a smaller generated beat drive, but their experiments did not match local polarization at the mixer.[5] Phase-locking value (PLV) also depends on baseline and state. These comparisons therefore do not calibrate the detector coefficient; that requires the same readout across a local-polarization amplitude sweep.

5.5 Limits of the population description

Cascading ordinary synaptic filters cannot remedy poor access: they attenuate the carrier before it reaches a subsequent detector. The response also depends on the neuron model: Hodgkin–Huxley and Frankenhaeuser–Huxley models reproduce stimulus-evoked firing missed by the tested leaky, exponential, and adaptive-exponential integrate-and-fire models [61], without invalidating

[5] The two-pair TI comparison gave a phase-locking effect ratio of 0.34 against tACS (95% confidence interval 0.04–0.65, 65 of 154 neurons), without correcting for TI currents up to ±2.5 mA per pair versus ±1 mA for tACS. A separate single-pair AM-tACS control compensated for frequency-dependent shunting to match extracellular modulation depth. Of twenty cells, ten responded to either modality and seven to both; in the latter group, median phase-locking value (PLV) was 0.10 for AM-tACS and 0.20 for tACS, against 0.043 at baseline. The reported TI modulation depths were about 0.5–0.62 V/m.

the weak-input expansion. Noise, heterogeneity beyond the assumed excitability distribution, and intercolumn interactions can broaden or shift the deterministic resonances. Human evidence is mixed, including peripheral effects without central effects [62] and null occipito-parietal alpha-power modulation [63]. Testing this mechanism therefore requires access, generated beat, and network tuning measured in the same preparation.

5.6 Falsifiable predictions and tests

Table 3 separates tests of access, detection, and amplification from observations already consistent with the model. Quantitative reanalysis of the timing and synaptic-blockade data requires the amplitude and harmonic controls of §3.2, with detector and network changes measured separately.

Network filtering must be separated from the harmonic signature. At resonance with $\mathcal{Q} = 10$, a detector ratio of 20% appears as only 0.7%, which could make an envelope follower look quadratic. Driving off resonance at $\Delta f \simeq f_0/4$ places both harmonics on the flatter part of the response, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ for $\mathcal{Q} = 3$–30. Alternatively, a weak direct tACS sweep can measure G independently so its effect can be removed from the harmonic ratio.

Table 3: Predictions, existing evidence, and tests. Published observations provide qualitative comparisons; their quantitative reanalysis remains to be done.

Prediction (model basis)	**Existing evidence**	**Proposed test**
Carrier dependence. At fixed local polarization, the leading beat is carrier-independent in the static limit and fast-carrier QIF asymptote, provided no other circuit mode is excited. Access restores dependence at fixed applied field; detector dynamics can add it near the crossover.	Carrier-frequency comparisons show changes in network sensitivity [6] and subthreshold single-cell mixing [10], without holding measured local carrier polarizations fixed. Thresholds depend on f_c [21, 22], and bare kHz fields have limited effects on hippocampal oscillations [64].	Measure each local polarization while sweeping f_c at fixed beat and state; normalize the weak beat by their product. Hold network gain fixed or measure it independently to separate access from residual detector dependence. Measure any residual cutoff rather than assigning it to a compartment.
Independent carrier amplitudes. At fixed detector and network gains, quadratic modulation excursion follows $2\lvert\varepsilon_1\varepsilon_2\rvert$; an ideal envelope follower tracks $2\min(\lvert\varepsilon_1\rvert, \lvert\varepsilon_2\rvert)$ (Eq. (9), Table 1).	The unequal-amplitude spatial example separates these maps; a biological test with independent local-polarization calibration is not supplied here.	Hold the weaker polarization fixed and vary the stronger, then swap tones. Keep both inputs weak, frequencies fixed, and the operating state controlled. Reconstruct the slow-response excursion after correcting network filtering; a beat lock-in alone does not measure the envelope's full excursion.

Continued on next page

Table 3 continued

Prediction (model basis)	Existing evidence	Proposed test
Current-ratio steering. With fixed electrode pairs, frequencies, access, and neural state, the quadratic beat map scales with $I_1^{\text{pair}} I_2^{\text{pair}}$; its normalized pattern does not move. An envelope follower can instead shift its maximum.	Cassarà et al. identify the product-law restriction [30]. The IXI002 comparison (§4.4) confirms the geometric distinction: envelope and mean-square foci move oppositely; the quadratic beat map only rescales. A physiological beat-map test remains open.	Vary the current ratio at fixed electrodes and compare normalized beat maps and target-to-reference ratios, using a cortex-only reference to test deep-target selectivity. Keep the projection direction fixed at each site, including any direction chosen to maximize the envelope at a reference ratio. Measure the DC shift and repeat at lower amplitude to check state dependence. Changing from $2+2$ to $1+3$ mA multiplies envelope excursion by 0.5–1.5 depending on the local polarization ratio, and the quadratic beat by 0.75 everywhere. The mean-square factor is 1.25 at equal per-current magnitudes, tending to 0.25 or 2.25 where the first or second carrier dominates, respectively. This last factor weights the signed DC source; it does not specify excitation or suppression.
State and carrier dependence. The static sigmoid follows σ'' and reverses sign at its inflection (PEIX$= 0$). Fast-carrier QIF mixes in voltage and reads the source through rate and network susceptibility; its DC term follows F' without the same sign rule (§2.9).	Cell-type-specific responses (PV versus pyramidal) are reported [16]; the detector coefficient remains model-dependent.	Vary tonic drive and carrier frequency at fixed beat; measure beat amplitude and phase. Compare with the predicted coefficient after accounting for network susceptibility. Amplitude changes alone do not isolate detection.
Transduction class. At a matched site, weak smooth detection with nonzero curvature gives $Y_\Omega \propto \varepsilon^2$ and a harmonic ratio vanishing as $\varepsilon \to 0$. An ideal envelope follower gives $Y_\Omega \propto \varepsilon$ and ratio 1/5; a threshold has onset $2\varepsilon = \delta$ and a variable ratio.	Untested with local-polarization calibration; existing comparisons match extracellular field rather than ε.	Sweep amplitude off resonance, reading both the log–log slope and $\lvert Y_{2\Omega}/Y_\Omega \rvert$. Near resonance, suppression by approximately $3\mathcal{Q}$ requires correction with an independently measured $G(2\Omega)/G(\Omega)$ (§3.2).
Frequency selectivity. The beat response peaks at the region's resonant rhythm and grows as $1/\gamma$ near its Hopf; state changes can alter both the generated drive and the recurrent gain [24].	The quantitative gain–bandwidth relation is not tested here in biological data.	Sweep Δf through the intrinsic band across brain states and track the peak frequency, height, and width.
Timing versus rate. Beat modulation can exceed the mean shift; their ratio peaks and then falls as the amplified oscillation rectifies (§4.6–4.9).	Qualitatively consistent with TI altering spike timing more than mean rate in primates [20].	Compare beat lock-in with mean-rate change in commensurate effect measures, retaining the finite-amplitude qualification.

Continued on next page

Table 3 continued

Prediction (model basis)	Existing evidence	Proposed test
Network readout. Population nonlinearity and synaptic recurrence contribute to the measured response.	Most pyramidal TI responses vanish under synaptic blockade [16], supporting network involvement without excluding cellular mixing.	Test whether synaptic blockade reduces recurrent amplification while preserving a cellular detector.
Equal-power carrier control. At second order, equal local RMS polarization gives the same steady source when the carriers see the same DC coefficient. TI also supplies a beat; either waveform can induce an oscillation if the mean shift crosses a bifurcation.	Demonstrated in the PING mean field (Fig. 13c,d). The corresponding spiking-network and in vivo controls are not provided here.	Compare two tones with one carrier of matched RMS polarization and duration, checking the DC coefficient across the carrier interval. Measure mean activity, intrinsic oscillations, and beat lock-in separately to distinguish the steady shift from the beat response.

6 Conclusion

TI requires access, detection, and amplification. For weak inputs with a nonzero second-order coefficient, detection generates a beat proportional to the product of the local carrier polarizations and a steady source proportional to their summed squares. Fast axonal access can be represented by a short effective membrane time, with its transfer still requiring calibration.

The detector law changes the spatial prediction. In the realistic-head model, changing the current ratio moves the envelope and mean-square foci in opposite directions but only rescales the quadratic beat at fixed neural state. Under uniform conductivity, access, and detection, the steady-source magnitude has no strict interior maximum.

Near a Hopf bifurcation, the circuit amplifies the beat in inverse proportion to modal damping. TI shares this linear amplifier with matched direct low-frequency stimulation. In the pinned JR sweep, the response is expressed mainly as beat-locked timing; its ratio to the mean shift eventually falls as the oscillation rectifies. At larger fields, the steady shift alone can induce gamma, as can an equal-power unmodulated carrier.

Local-polarization measurements, independent-amplitude tests, beat sweeps across states, and equal-power controls separate these mechanisms. Together they test whether a measured TI response reflects access, nonlinear detection, recurrent gain, or a field-induced change of state.

Code and data availability

Simulation code, cached results, and figure-generation scripts are provided at the paper's GitHub repository; the public archive is linked through Zenodo (doi:10.5281/zenodo.21009618). The repository `README` maps figures to their scripts and distinguishes recomputation from replotting cached data. The two-tone kHz example and both LaNMM carrier-measurement windows have cached outputs and reproduction scripts. The FEM producer `code/fem_quadmaps.py` computes the reported maps and statistics from the stored fields; the summary arrays and `code/make_fem_fig.py` reproduce Fig. 8 without the head model. Recomputing its maps from the underlying fields additionally requires Neuroelectrics' nonpublic temporal-interference library and the stored IXI002 fields and labels; the input hashes and settings are recorded in `docs/fem_ixi002_violante.md` and `code/fem_quadmaps.json`. The code in this paper's repository is released under CC-BY-4.0.

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Competing interests

Giulio Ruffini, Borja Mercadal, Ricardo Salvador, Alex Just, and Francesca Castaldo are employees of Neuroelectrics, which develops and sells transcranial electrical stimulation devices. Giulio Ruffini is a co-founder and shareholder of Neuroelectrics. The other authors declare no competing interests.

Author contributions

Using the CRediT taxonomy: **G. Ruffini**: conceptualization, methodology, formal analysis, software, investigation, visualization, writing (original draft), supervision, funding acquisition. **B. Mercadal**: validation, visualization, writing (review and editing). **R. Salvador**: conceptualization (current-ratio steering), methodology (head modeling), software, formal analysis, data curation, visualization. **A. Just**: software, visualization, writing (review and editing). **R. de Palma Aristides**: software, visualization, writing (review and editing). **F. Castaldo**: conceptualization, writing (review and editing). **N. Grossman**: conceptualization. **S. Canals**: supervision, writing (review and editing). **C. Mirasso**: supervision, writing (review and editing).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

References

[1] Nir Grossman, David Bono, Nina Dedic, Suhasa B. Kodandaramaiah, Andrii Rudenko, Ho-Jun Suk, Antonino M. Cassara, Esra Neufeld, Niels Kuster, Li-Huei Tsai, Alvaro Pascual-Leone, and Edward S. Boyden. Noninvasive deep brain stimulation via temporally interfering electric fields. *Cell*, 169(6):1029–1041.e16, 2017.

[2] Ehsan Mirzakhalili, Beatrice Barra, Marco Capogrosso, and Scott F. Lempka. Biophysics of temporal interference stimulation. *Cell Systems*, 11(6):557–572.e5, 2020.

[3] Paul J. Nahin. *The Science of Radio: with MATLAB and Electronics Workbench Demonstrations*. Springer, 2nd edition, 2001.

[4] Yair Deitcher, Guy Eyal, Lida Kanari, Matthijs B. Verhoog, Guy Antoine Atenekeng Kahou, Huibert D. Mansvelder, Christiaan P. J. de Kock, and Idan Segev. Comprehensive morpho-electrotonic analysis shows 2 distinct classes of L2 and L3 pyramidal neurons in human temporal cortex. *Cerebral Cortex*, 27(11):5398–5414, 2017.

[5] Guy Eyal, Matthijs B. Verhoog, Guilherme Testa-Silva, Yair Deitcher, Johannes C. Lodder, Ruth Benavides-Piccione, Juan Morales, Javier DeFelipe, Christiaan P. J. de Kock, Huibert D. Mansvelder, and Idan Segev. Unique membrane properties and enhanced signal processing in human neocortical neurons. *eLife*, 5:e16553, 2016.

[6] Zeinab Esmaeilpour, Greg Kronberg, Davide Reato, Lucas C. Parra, and Marom Bikson. Temporal interference stimulation targets deep brain regions by modulating neural oscillations. *Brain Stimulation*, 14(1):55–65, 2021.

[7] Charles C. H. Cohen, Marko A. Popovic, Jan Klooster, Marie-Theres Weil, Wiebke Möbius, Klaus-Armin Nave, and Maarten H. P. Kole. Saltatory conduction along myelinated axons involves a periaxonal nanocircuit. *Cell*, 180(2):311–322.e15, 2020.

[8] Maarten H. P. Kole, Susanne U. Ilschner, Björn M. Kampa, Stephen R. Williams, Peter C. Ruben, and Greg J. Stuart. Action potential generation requires a high sodium channel density in the axon initial segment. *Nature Neuroscience*, 11(2):178–186, 2008.

[9] Boshuo Wang, Aman S. Aberra, Warren M. Grill, and Angel V. Peterchev. Responses of model cortical neurons to temporal interference stimulation and related transcranial alternating current stimulation modalities. *Journal of Neural Engineering*, 19(6):066047, 2022.

[10] Charlotte E. Luff, Robert Peach, Emma-Jane Mallas, Edward Rhodes, Felix Laumann, Edward S. Boyden, David J. Sharp, Mauricio Barahona, and Nir Grossman. The neuron mixer and its impact on human brain dynamics. *Cell Reports*, 43(6):114274, 2024.

[11] Bruce Hutcheon and Yosef Yarom. Resonance, oscillation and the intrinsic frequency preferences of neurons. *Trends in Neurosciences*, 23(5):216–222, 2000.

[12] Raul de Palma Aristides, Leonardo L. Gollo, Francesca Castaldo, Giulio Ruffini, Santiago Canals, and Claudio R. Mirasso. Frequency-selective responses to temporal interference stimulation in hippocampal neurons. Unpublished manuscript, September 29, 2026. BCOM Working Paper WP0242, 2026.

[13] Nicolas Brunel and Vincent Hakim. Fast global oscillations in networks of integrate-and-fire neurons with low firing rates. *Neural Computation*, 11(7):1621–1671, 1999.

[14] Nicolas Brunel and Xiao-Jing Wang. What determines the frequency of fast network oscillations with irregular neural discharges? I. synaptic dynamics and excitation–inhibition balance. *Journal of Neurophysiology*, 90(1):415–430, 2003.

[15] Erwan Ledoux and Nicolas Brunel. Dynamics of networks of excitatory and inhibitory neurons in response to time-dependent inputs. *Frontiers in Computational Neuroscience*, 5, 2011.

[16] Sara Caldas-Martinez, Chaitanya Goswami, Mats Forssell, Jiaming Cao, Alison L. Barth, and Pulkit Grover. Cell-specific effects of temporal interference stimulation on cortical function. *Communications Biology*, 7:1076, 2024.

[17] Nafiseh Karimi, Rassoul Amirfattahi, and Abolghasem Zeidaabadi Nezhad. Neuromodulation effect of temporal interference stimulation based on network computational model. *Frontiers in Human Neuroscience*, 18, September 2024.

[18] Hossein Soroushi, Sara Abbasi, Yun Du, Ning Ning, and Yuming Lei. Temporal interference stimulation: mechanisms, optimization, validation, and clinical prospects—a comprehensive review. *WIREs Computational Statistics*, 17(3):e70031, 2025.

[19] Wenjiang Wang, Florian Missey, Chenxi Ling, Kun Wei, Junhao Jian, Haoxuan Sun, Liujiao Yan, Shaohua Hu, and Bo Bi. Temporal interference stimulation: a new paradigm for non-invasive deep brain stimulation—mechanistic insights, clinical advances, and future directions. *Translational Psychiatry*, 2026.

[20] Pedro G. Vieira, Matthew R. Krause, and Christopher C. Pack. Temporal interference stimulation disrupts spike timing in the primate brain. *Nature Communications*, 15:4558, 2024.

[21] Aleksandar Opančar, Petra Ondráčková, David Samuel Rose, Jan Trajlinek, Vedran Đerek, and Eric Daniel Głowacki. The same biophysical mechanism is involved in both temporal

interference and direct kHz stimulation of peripheral nerves. *Nature Communications*, 16:9006, 2025.

[22] Ryan B. Budde, Michael T. Williams, and Pedro P. Irazoqui. Temporal interference current stimulation in peripheral nerves is not driven by envelope extraction. *Journal of Neural Engineering*, 20(2):026041, 2023.

[23] Iurii Semenov, Vitalii Kim, Giedre Silkuniene, and Andrei G. Pakhomov. Excitation and polarization of isolated neurons by high-frequency sine waves for temporal interference stimulation. *Cell Reports Physical Science*, 6(7):102660, 2025.

[24] Giulio Ruffini, Edmundo Lopez-Sola, Raul Palma, Roser Sanchez-Todo, Jakub Vohryzek, Francesca Castaldo, and Karl Friston. Cross-frequency coupling as a neural substrate for prediction error evaluation: a laminar neural mass modeling approach. *Neural Computation*, 38(8):1299–1343, 2026.

[25] Giulio Ruffini. Neural encoding through hierarchical amplitude modulation. *bioRxiv*, 2025. Preprint 2025.11.03.686310.

[26] Francesca Castaldo, Raul de Palma Aristides, Pau Clusella, Jordi Garcia-Ojalvo, and Giulio Ruffini. Rosetta Stone of neural mass models. *Physics Reports*, 1189:1–49, 2026. arXiv:2512.10982.

[27] Ben H. Jansen and Vincent G. Rit. Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366, 1995.

[28] Hugh R. Wilson and Jack D. Cowan. Excitatory and inhibitory interactions in localized populations of model neurons. *Biophysical Journal*, 12(1):1–24, 1972.

[29] Adrià Galan-Gadea, Ricardo Salvador, Fabrice Bartolomei, Fabrice Wendling, and Giulio Ruffini. Spherical harmonics representation of the steady-state membrane potential shift induced by tDCS in realistic neuron models. *Journal of Neural Engineering*, 20(2):026004, 2023.

[30] Antonino M. Cassarà, Taylor H. Newton, Katie Zhuang, Sabine J. Regel, Peter Achermann, Alvaro Pascual-Leone, Niels Kuster, and Esra Neufeld. Recommendations for the safe application of temporal interference stimulation in the human brain part II: Biophysics, dosimetry, and safety recommendations. *Bioelectromagnetics*, 46(1):e22536, 2025.

[31] Ernest Montbrió, Diego Pazó, and Alex Roxin. Macroscopic description for networks of spiking neurons. *Physical Review X*, 5(2):021028, 2015.

[32] Giulio Ruffini. Analysis and extension of exact mean-field theory with dynamic synaptic currents. *bioRxiv*, 2021. Preprint 2021.09.01.458563 (NMM2).

[33] Pau Clusella, Elif Köksal-Ersöz, Jordi Garcia-Ojalvo, and Giulio Ruffini. Comparison between an exact and a heuristic neural mass model with second-order synapses. *Biological Cybernetics*, 117(1-2):5–19, 2023.

[34] Jesper D. Nielsen, Kristoffer H. Madsen, Oula Puonti, Hartwig R. Siebner, Christian Bauer, Camilla Gøbel Madsen, Guilherme B. Saturnino, and Axel Thielscher. Automatic skull segmentation from MR images for realistic volume conductor models of the head: assessment of the state-of-the-art. *NeuroImage*, 174:587–598, 2018.

[35] Christian Gaser, Robert Dahnke, Paul M. Thompson, Florian Kurth, Eileen Luders, and Alzheimer's Disease Neuroimaging Initiative. CAT: a computational anatomy toolbox for the analysis of structural MRI data. *GigaScience*, 13:giae049, 2024.

[36] Ricardo Salvador, Junhong Zhou, Brad Manor, and Giulio Ruffini. Group optimization methods for dose planning in tES. *Journal of Neural Engineering*, 22:046045, 2025.

[37] Ines R. Violante, Ketevan Alania, Antonino M. Cassarà, Esra Neufeld, Emma Acerbo, Romain Carron, Adam Williamson, Danielle L. Kurtin, Edward Rhodes, Adam Hampshire, Niels Kuster, Edward S. Boyden, Alvaro Pascual-Leone, and Nir Grossman. Non-invasive temporal interference electrical stimulation of the human hippocampus. *Nature Neuroscience*, 26(11):1994–2004, 2023.

[38] K. Yatsuda, W. Yu, and J. Gomez-Tames. Population-level insights into temporal interference for focused deep brain neuromodulation. *Frontiers in Human Neuroscience*, 18:1308549, 2024.

[39] O. C. Zienkiewicz and J. Z. Zhu. The superconvergent patch recovery and a posteriori error estimates. Part 1: The recovery technique. *International Journal for Numerical Methods in Engineering*, 33(7):1331–1364, 1992.

[40] Thomas Radman, Raddy L. Ramos, Joshua C. Brumberg, and Marom Bikson. Role of cortical cell type and morphology in subthreshold and suprathreshold uniform electric field stimulation in vitro. *Brain Stimulation*, 2(4):215–228.e3, October 2009.

[41] Marom Bikson, Masashi Inoue, Hiroki Akiyama, Jackie K. Deans, John E. Fox, Hiroyoshi Miyakawa, and John G. R. Jefferys. Effects of uniform extracellular dc electric fields on excitability in rat hippocampal slices in vitro. *The Journal of Physiology*, 557(1):175–190, May 2004.

[42] Jacqueline K. Deans, Andrew D. Powell, and John G. R. Jefferys. Sensitivity of coherent oscillations in rat hippocampus to ac electric fields. *The Journal of Physiology*, 583(2):555–565, August 2007.

[43] Bruce Fischl, David H. Salat, Evelina Busa, Marilyn Albert, Megan Dieterich, Christian Haselgrove, Andre van der Kouwe, Ron Killiany, David Kennedy, Shuna Klaveness, Albert Montillo, Nikos Makris, Bruce Rosen, and Anders M. Dale. Whole brain segmentation: automated labeling of neuroanatomical structures in the human brain. *Neuron*, 33(3):341–355, 2002.

[44] François Grimbert and Olivier Faugeras. Bifurcation analysis of Jansen's neural mass model. *Neural Computation*, 18(12):3052–3068, 2006.

[45] Martin Vinck, Marijn van Wingerden, Thilo Womelsdorf, Pascal Fries, and Cyriel M. A. Pennartz. The pairwise phase consistency: a bias-free measure of rhythmic neuronal synchronization. *NeuroImage*, 51(1):112–122, 2010.

[46] Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.

[47] Ryan T. Canolty and Robert T. Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010.

[48] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010.

[49] Michal Cifra, Saurabh Kumar Pandey, Tomáš Zakar, Michaela Poplová, Daniela G. Blanco Campoy, Jiří Průša, Daniel Havelka, Paolo Marracino, Micaela Liberti, Francesca Apollonio, Niall J. English, Carl Caleman, and Erik G. Marklund. The effects of external electric fields on proteins. *Chemical Society Reviews*, 55(14):7568–7616, 2026.

[50] Asif Rahman, Davide Reato, Mattia Arlotti, Fernando Gasca, Abhishek Datta, Lucas C. Parra, and Marom Bikson. Cellular effects of acute direct current stimulation: somatic and synaptic terminal effects. *The Journal of Physiology*, 591(10):2563–2578, 2013.

[51] Aman S. Aberra, Ruochen Wang, Warren M. Grill, and Angel V. Peterchev. Multi-scale model of axonal and dendritic polarization by transcranial direct current stimulation in realistic head geometry. *Brain Stimulation*, 16(6):1776–1791, 2023.

[52] Darpan Chakraborty, Dennis Q. Truong, Marom Bikson, and Hanoch Kaphzan. Neuromodulation of axon terminals. *Cerebral Cortex*, 28(8):2786–2794, 2018.

[53] Charlotte E. Luff, Patrycja Dzialecka, Emma Acerbo, Adam Williamson, and Nir Grossman. Pulse-width modulated temporal interference (pwm-ti) brain stimulation. *Brain Stimulation*, 17(1):92–103, January 2024.

[54] Florian Missey, Emma Acerbo, Adam S. Dickey, Jan Trajlinek, Ondřej Studnička, Claudia Lubrano, Mariane de Araújo e Silva, Evan Brady, Vit Všianský, Johanna Szabo, Irena Dolezalova, Daniel Fabo, Martin Pail, Claire-Anne Gutekunst, Rosanna Migliore, Michele Migliore, Stanislas Lagarde, Romain Carron, Fariba Karimi, Raul Castillo Astorga, Antonino M. Cassara, Niels Kuster, Esra Neufeld, Fabrice Bartolomei, Nigel P. Pedersen, Robert E. Gross, Viktor Jirsa, Daniel L. Drane, Milan Brázdil, and Adam Williamson. Non-invasive temporal interference stimulation of the hippocampus suppresses epileptic biomarkers in patients with epilepsy: biophysical differences between kilohertz and amplitude modulated stimulation. *Brain Stimulation*, 19(1):102981, 2026.

[55] M. M. Ali, K. K. Sellers, and F. Frohlich. Transcranial alternating current stimulation modulates large-scale cortical network activity by network resonance. *Journal of Neuroscience*, 33(27):11262–11275, July 2013.

[56] Ehsan Negahbani, Florian H. Kasten, Christoph S. Herrmann, and Flavio Fröhlich. Targeting alpha-band oscillations in a cortical model with amplitude-modulated high-frequency transcranial electric stimulation. *NeuroImage*, 173:3–12, June 2018.

[57] Matthew R. Krause, Pedro G. Vieira, and Christopher C. Pack. Transcranial electrical stimulation: How can a simple conductor orchestrate complex brain activity? *PLOS Biology*, 21(1):e3001973, January 2023.

[58] Jens Wilting and Viola Priesemann. Inferring collective dynamical states from widely unobserved systems. *Nature Communications*, 9(1), June 2018.

[59] Yu Huang, Anli A Liu, Belen Lafon, Daniel Friedman, Michael Dayan, Xiuyuan Wang, Marom Bikson, Werner K Doyle, Orrin Devinsky, and Lucas C Parra. Measurements and models of electric fields in the in vivo human brain during transcranial electric stimulation. *eLife*, 6, February 2017.

[60] Yu Huang, Anli A Liu, Belen Lafon, Daniel Friedman, Michael Dayan, Xiuyuan Wang, Marom Bikson, Werner K Doyle, Orrin Devinsky, and Lucas C Parra. Correction: measurements and models of electric fields in the in vivo human brain during transcranial electric stimulation. *eLife*, 7, February 2018.

[61] Tom Plovie, Ruben Schoeters, Thomas Tarnaud, Wout Joseph, and Emmeric Tanghe. Nonlinearities and timescales in neural models of temporal interference stimulation. *Bioelectromagnetics*, 46(1):e22522, 2025.

[62] Krisztián Iszak, Simon Mathies Gronemann, Stefanie Meyer, Alexander Hunold, Jana Zschüntzsch, Mathias Bähr, Walter Paulus, and Andrea Antal. Why temporal inference stimulation may fail in the human brain: A pilot research study. *Biomedicines*, 11(7):1813, June 2023.

[63] Paria Mansourinezhad, Rob M C Mestrom, Debby C W Klooster, Mathieu Sprengers, Paul A J M Boon, and Margarethus M Paulides. Systematic review of experimental studies in humans on transcranial temporal interference stimulation. *Journal of Neural Engineering*, 22(5):051001, September 2025.

[64] Zeinab Esmaeilpour, Mark Jackson, Greg Kronberg, Tianhe Zhang, Rosana Esteller, Brad Hershey, and Marom Bikson. Limited sensitivity of hippocampal synaptic function or network oscillations to unmodulated kilohertz electric fields. *eNeuro*, 7(6):ENEURO.0368–20.2020, 2020.

A Notation and units

Table 4 groups the symbols by their role in the calculation. JR and LaNMM voltages are in mV and rates in s^{-1}. QIF voltages and current-like drives are dimensionless model variables; QIF rates are physical rates, integrated in ms^{-1} and converted to Hz where indicated. The uncoupled MPR derivation explicitly uses membrane-time units; the main population equations retain physical time and rate. All frequencies and time constants in a formula must use the same time unit. In the FEM comparison $\mathbf{n}^*$ is the direction maximizing the envelope at balanced currents, held fixed during the ratio sweep. A star otherwise marks a fixed point or, on a bifurcation parameter, its critical value. Subscripts E and I identify excitatory and inhibitory populations; they are not field or current variables.

Table 4: Notation and units. "Normalized" means dimensionless in the stated model scaling; output units depend on the named observable. Numbered references identify main-text definitions; lettered locations identify appendices (in the supplement for the JNE edition).

Symbol	Meaning	Units	Defined in
Field and access			
$E(t), E_0, \mathbf{E}_j$	Applied field, its amplitude, and vector field of tone $j = 1, 2$	V/m	Eq. (3)
$s(t), \varepsilon_j, \varepsilon$	Local polarization; tone amplitudes; common amplitude ($\varepsilon_{\mathrm{TI}}$ and $\varepsilon_{\mathrm{tACS}}$ label the waveform)	mV	Eqs. (1), (4)
$I_j^{\mathrm{pair}}, \mathcal{P}_j$	Peak electrode-pair current; signed local polarization per unit current	mA; mV/mA	§2.3
$\mathbf{F}_j, \sigma_{\mathrm{cond}}$	FEM field per pair current; tissue conductivity	(V/m)/mA; S/m	§3.3
$\mathcal{A}(t), \overline{\mathcal{A}}$	Nonnegative polarization envelope and its beat-period mean	mV	Eq. (8); Table 1
f_1, f_2, f_c	Tone frequencies and their center	Hz	Eq. (1)
ω_j, ω_c	Angular frequencies $2\pi f_j, 2\pi f_c$	rad/s	Eqs. (1), (4)
$\Delta f, \Omega$	Beat frequency $f_2 - f_1$; $\Omega = 2\pi\Delta f$	Hz; rad/s	Eq. (1)
m	AM-surrogate modulation depth	dimensionless	Eq. (20)
$\Lambda[E], \lambda_0, \lambda_f$	Field-induced polarization; static and frequency-dependent coupling	mV; mV/(V/m)	Eqs. (1), (4)
$\boldsymbol{\lambda}_0^k, \boldsymbol{\lambda}_f^k$	Coupling vectors for compartment k	mV/(V/m)	Eq. (4)
H, H_k, H_m	Dominant-compartment access transfer, indexed transfer, and point-membrane transfer	dimensionless	Fig. 3; §2.2, §2.7
τ_m, τ	Membrane time; access-filter time in kHz controls	ms	Eqs. (17), (15)
f_{char}	Characteristic frequency $1/(2\pi\tau)$	Hz	Table 2
$\ell(\omega), \ell_0$	Frequency-dependent and static cable space constants	μm	App. B.1, Eq. (5)
$\mathbf{V}, \mathsf{C}_m, \mathsf{G}_m$	Compartment voltages, capacitance and conductance matrices	V; F; S	App. B.1
$\mathbf{I}^0, \mathbf{p}_E$	Endogenous currents; field-to-current vector	A; A/(V/m)	App. B.1
$\mathbf{x}, \phi_j, \mathbf{n}$	Position, field potential, unit cell orientation	m; V; unit vector	§2.3, §2.5
$\mathcal{I}$	Nonnegative mean-square local polarization before detector gain	mV^2	§2.5
Detection			
$\sigma, \sigma', \sigma''$	JR/LaNMM rate curve, slope and curvature	s^{-1}; $s^{-1}mV^{-1}$; $s^{-1}mV^{-2}$	Eq. (2)
e_0, v_0, ρ	Half the rate ceiling, inflection voltage, inverse voltage scale	s^{-1}; mV; mV^{-1}	Eq. (2)
PEIX	Signed sigmoid saturation index	dimensionless	Eq. (16)
$Y_\Omega, Y_{\mathrm{DC}}$	Complex beat amplitude and signed steady source before recurrence	detector-output units	Eqs. (6), (5)

Continued on next page

Table 4 continued

Symbol	Meaning	Units	Defined in
$\chi^{(2)}$	Dynamic coefficient mapping a polarization product to a rate beat	$s^{-1}mV^{-2}$	§2.2
g, T, Y_{LF}	Memoryless response, envelope transduction, carrier-averaged output	specified output units	Eq. (11)
c_1, c_2, c_4	Linear, quadratic and quartic envelope coefficients	output/$(\mathrm{mV})^{1,2,4}$	§2.4
$\mathcal{H}_{\mathrm{det}}, \mathcal{H}_{\mathrm{meas}}$	Second/fundamental harmonic magnitude ratios before and after the network	dimensionless	Eq. (12)
$\mathcal{M}[y]$	Peak-to-peak excursion of the quantity y	units of y	Eq. (9)
$F(Q), F', F''$	MPR steady-rate curve and drive derivatives	normalized	Eq. (18)
Q, q	Mean total drive $\bar{\eta} + I$; single-cell drive $\eta + I$	normalized	Eq. (18); App. B.3
I_{ac}	Single-tone current amplitude	normalized	§2.9
$\kappa_{\mathrm{DC}}, \kappa_V$	Mean-rate mixing per squared current or squared voltage excursion	normalized	Eqs. (6), (8)
d, ν, ω_z	MPR relaxation, ringing and zero-crossing angular frequencies	membrane-time units	App. B.2
$\mathsf{J}, \mathbf{h}, H_v$	Reduced MPR Jacobian, unit-current response and voltage transfer	normalized	§2.9; App. B.3
$\mathbf{b}_2, \mathsf{I}_2$	Quadratic DC source per I_{ac}^2; identity matrix	normalized	App. B.3
$\mathcal{R}_{\mathrm{det}}, \mathcal{R}_{\mathrm{lin}}$	TI/direct-drive source ratio; linear TI/tACS output ratio	dimensionless	App. D; §4.7
δ, ζ	Threshold offset; sigmoid rate divided by $2e_0$	mV; dimensionless	§2.4; App. D
$u, \xi, \beta, k_{\mathrm{ell}}^2$	Reduced amplitude $\rho\varepsilon$, offset $\rho(v^* - v_0)$, threshold ratio, elliptic parameter	dimensionless	App. D
$L, \psi, \mathsf{K}, \mathsf{E}$	Unit logistic, normalized waveform, complete elliptic integrals	dimensionless	App. D
Network			
r_X, v_X	Population firing rate and mean potential, indexed by X	s^{-1}; mV in JR/LaNMM, normalized in QIF	§2.1
t, r, f (MPR derivation)	Time, population rate and single-cell rate in the stated membrane-time convention	normalized locally	App. B.2
$r_{\mathrm{syn}}, r_E^{\mathrm{syn}}, r_I^{\mathrm{syn}}$	Synaptically filtered firing rates	same rate units as input	Eqs. (19), (22)
$u_{\mathrm{PSP}}, u_{X\leftarrow Y}, \hat{L}$	Postsynaptic potential and inverse synaptic operator	mV; maps mV to s^{-1}	Eq. (1)
A, B, a, b	JR excitatory/inhibitory PSP gains and inverse synaptic times	mV; s^{-1}	§2.1; App. B
h_e, h_i	JR excitatory and inhibitory synaptic impulse responses	mV	Fig. 3
$C, C_{1\ldots4}, C_{X\leftarrow Y}$	JR connectivity scale and pathway weights	dimensionless	§2.1; App. B
y_0, y_1, y_2, v	JR PSPs; voltage readout $v = y_1 - y_2$	mV	App. B
p, p_H, C^*	JR external rate and its upper Hopf value; critical connectivity	s^{-1}; dimensionless for C^*	§3.4, §4.5
$\eta_j, \bar{\eta}, \Delta$	QIF excitabilities, Lorentzian center and half-width	normalized	Eq. (17)
$I, I_E, I_j, A_f, \lambda_I$	QIF input, E-population input, tone amplitudes, forcing amplitude and field coupling	normalized; λ_I in $(\mathrm{V/m})^{-1}$	Eqs. (17), (22)
$\mathcal{U}_j$	Total synaptic and field drive to QIF neuron j	normalized	Fig. 3
J, J^*, w_{XY}	QIF coupling, critical coupling and relative pathway weights	dimensionless	Eq. (22)

Continued on next page

Table 4 continued

Symbol	Meaning	Units	Defined in
$\tau_E, \tau_I, \tau_a, \tau_g, \tau_s$	E/I membrane and excitatory/inhibitory/generic synaptic times	ms	Eqs. (19), (22)
$\hat{K}_a, \hat{K}_g, H_s$	QIF synaptic rate filters; JR rate-to-voltage transfer	dimensionless; mV s for H_s	§3.5, §2.6
$\mu_\pm, \gamma, \omega_0, f_0$	Modal eigenvalues, damping, angular and ordinary modal frequency	s^{-1} (ms^{-1} where stated); Hz for f_0	Eq. (13)
$\tau_{\mathrm{rel}}, \tau_{\mathrm{rev}}, \mathcal{Q}$	Population relaxation time, $1/\gamma$, and mode quality factor	time; dimensionless for $\mathcal{Q}$	§2.1, §2.6, §5
D, G, R_Ω	Beat-source coefficient, recurrent transfer, complex observed beat response	input/output-dependent	Eq. (14)
$V_j, V_{\mathrm{peak}}, N$	Single-QIF voltage, spike cutoff and neurons per population	normalized; count	§3.6
$\delta\eta_E, \delta\eta_E^*$	Rectified E-drive shift and its Hopf threshold	normalized	§3.6, §4.9
P_1, P_2, r_P, φ_X	Deep/superficial pyramidal populations; JR and LaNMM pyramidal rates	labels; s^{-1} for rates	§4.6; §E.1
$z, \alpha, \mathcal{F}, \varepsilon_{\mathrm{SL}}$	Stuart–Landau mode, growth rate, forcing and forcing amplitude	normalized mode; s^{-1} for the others	App. C
Measurement			
$L_{\cos}, L_{\sin}, A_\Omega$	Signed lock-in quadratures and nonnegative amplitude	measured voltage or rate units	Eq. (21)
A_{open}	Magnitude of the open-loop rate beat	s^{-1}	§4.5
$w_k, t_k, v_k, \bar{v}$	Hann weights, sample times, voltage samples and weighted mean	1; s; mV; mV	Eq. (21)
$A_{\mathrm{test}}, \theta_0$	Amplitude and phase of the lock-in test tone	readout units; rad	§3.2
$\Delta t, N_{\mathrm{grid}}$	Integration step and number of parameter combinations	s or ms; count	§3.4
$A_\Omega/\langle r_E \rangle$	Beat-locked population modulation depth	dimensionless	§3.6
PPC	Pairwise phase consistency of single-neuron spike phases	dimensionless	§3.6
$f_{\mathrm{out}}, S(f_c)$	Dominant output frequency; LaNMM mean off-alpha lock-in	Hz; mV	App. F; §E.1